A fast upper envelope scan method for discrete-continuous dynamic programming*

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Abstract

We develop a general *fast upper envelope scan* (FUES) method for solving stochastic dynamic problems with discrete-continuous choices, including when standard envelope-based solution methods fail due to non-monotonic policy functions and the presence of multiple locally optimal policy candidates. Such failures arise because existing endogenous grid method (EGM) implementations rely on monotonicity assumptions that are frequently violated in economics, forcing researchers either to impose restrictive structures or revert to slower and less precise global solution methods. FUES overcomes these limitations by providing an upper-envelope operator that accommodates all policy mappings while retaining the computational advantages of EGM. Under standard conditions, we prove the method always recovers the correct upper envelope and, using benchmark applications, demonstrate large computational gains, thereby significantly expanding the class of discrete-continuous models amenable to efficient envelope-based solution methods.

Key Words: discrete-continuous choices, non-convex optimization, Euler equation, stochastic dynamic programming

JEL Classification: C13, C63, D91

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1 Introduction

Stochastic dynamic decision-making applications have increasingly incorporated discrete-continuous choices. Such choices are central to understanding the dynamics of investment frictions (Skiba, 1978; Khan and Thomas, 2008), housing adjustments (Yogo, 2016; Kaplan et al., 2020), durable goods consumption and macroeconomic dynamics (Kaplan and Violante, 2014; Berger and Vavra, 2015; Monninger, 2025), life cycle asset allocation (Fagereng et al., 2019; Maxted et al., 2024), labor supply (Attanasio et al., 2018; Iskhakov and Keane, 2021) and default risks (Arellano, 2008; Arellano et al., 2016; Jang and Lee, 2023), among many other areas of active research.

Solving these models efficiently remains challenging because discrete-continuous choice problems typically generate non-concave value functions and non-monotone policy correspondences. Currently, the standard approach to solve stochastic dynamic problems is the endogenous grid method (EGM). Unlike value function iteration (VFI), EGM relies on analytically inverting the Euler equation (Carroll, 2006; White, 2015; Lujan, 2025) and can eliminate costly numerical root-finding or optimization steps (Iskhakov, 2015). However, like other Euler-equation-based methods (Coleman, 1990; Li and Stachurski, 2014), EGM relies on concavity to ensure that Euler equation solutions are globally optimal. This concavity generally fails in models with discrete-continuous choices, leading to a policy correspondence rather than a single-valued policy function, with multiple locally optimal policy candidates at a given state (Fella, 2014; Iskhakov et al., 2017). Several methods have been proposed to recover the upper envelope of the solutions generated by the discrete-continuous EGM (DC-EGM). These methods rely either on strict monotonicity of the optimal policy function (Fella, 2014; Iskhakov et al., 2017; Druedahl and Jørgensen, 2017)[1] or on combining Euler equation methods with numerical optimization (Druedahl, 2021). They all, however, scale poorly with model complexity, lack any formal guarantees of recovering the correct upper envelope, and offer no error bounds for the resulting policy functions.

We address these limitations by introducing a general and efficient method for constructing the upper envelope of the EGM value correspondence in discrete-continuous problems that admit a one-dimensional decision state. Our fast upper envelope scan (FUES) method exploits the fact that the upper envelope of the value correspondence is the

[1]The monotonicity requirement of existing methods limits their applicability when discrete-continuous choices introduce both upward and downward jumps or flat inaction regions in the optimal continuation state policy function - as in models with durable and non-durable consumption (Kaplan and Violante, 2014; Yogo, 2016), present-bias (Laibson, 1997), or non-linear taxation (Best and Kleven, 2018; Gelber et al., 2020).

supremum of choice-specific concave value functions, each associated with a history-dependent future sequence of discrete choices. Non-concavities arise only at crossing points of these choice-specific value functions, where the optimal policy function exhibits discontinuous jumps. FUES identifies these non-concave regions using a targeted secant scan that determines whether including a potential optimal point forms a concave or convex region of the upper envelope, eliminating suboptimal candidates while retaining those that define convex regions. This process recovers the global upper envelope without imposing monotonicity or requiring numerical optimization.

We illustrate the applicability of FUES via three applications. First, we implement FUES in a classic discrete-continuous dynamic model - i.e., a life cycle model with discrete retirement and continuous consumption decisions à la Iskhakov et al. (2017). Policy function monotonicity is preserved here, making it a natural benchmark for evaluating speed and accuracy in applications with strictly monotone continuation state policies. Second, we show how FUES enables the EGM in a problem of continuous housing investment with adjustment frictions (Kaplan and Violante, 2014; Yogo, 2016; Dobrescu et al., 2025). Here, existing EGM implementations fail because the intertemporal income effects between housing investments and financial savings lead to multiple inverse Euler equation solutions and non-monotone housing policies. Finally, we apply FUES to an extended version of the discrete housing choice model in Fella (2014) that expands the set of discrete options and brings the decision structure closer to the multi-stage discrete-continuous problems encountered in larger applied models. In this case, existing EGM implementations fail because the (non-smooth) capital income tax schedule leads to behavioral inaction regions, breaking down the strict monotonicity of the savings policy.

For each application, we compare FUES to the applicable solution methods, including the monotone segment selection (MSS) in Iskhakov et al. (2017), the local triangulation (LTM) in Druedahl and Jørgensen (2017) and Druedahl (2021), the multidimensional rooftop cut method (RFC) in Dobrescu and Shanker (2024), the nested-EGM (NEGM) in Druedahl (2021), and VFI. In all cases, FUES delivers sizable reductions in computation time, particularly as grid sizes increase and the cost of envelope construction becomes a binding constraint. For instance, FUES yields better total solution time by up to one order of magnitude relative to MSS, by more than two orders of magnitude relative to LTM, and by one to two orders of magnitude compared to RFC. These gains are sometimes less pronounced in environments where monotonicity fails and alternative methods rely on numerical optimization or grid refinement but even in this case, FUES delivers better total solution time by about 25-50% relative to NEGM, and two orders of magnitude than VFI.

While computational speed matters greatly for the practical applicability of a method, the solution accuracy it delivers is paramount. Whenever MSS or LTM are applicable, their Euler equation errors are essentially identical to those in FUES and RFC, as all evaluate the upper envelope of EGM candidate points without numerical optimization. When monotonicity fails and MSS and LTM are unusable, FUES continues to recover the correct envelope while avoiding the accuracy losses associated with numerical optimization-based approaches such as NEGM and VFI. In these cases, FUES achieves accuracy levels comparable with RFC but much better than NEGM and VFI (i.e., two to 2.5 orders of magnitude lower Euler equation errors) at substantially lower computational cost.

Finally, we provide the theoretical foundation for FUES. In doing so, we show how it can recover the upper envelope of the approximate value functions that form the EGM value correspondence without error when (i) the jumps between policy functions are bounded below (as is the case when there are finitely many jumps in the policy functions), (ii) the grid size is sufficiently large, and (iii) the smooth regions of the problem satisfy standard economic conditions: concavity of the value functions and Lipschitz continuity of the policy functions - all straightforward to verify in practice.

Relative to existing methods, FUES makes three contributions. First, by eliminating the requirement of policy function monotonicity, it makes envelope-based solution methods applicable to discrete-continuous models with uncertainty and non-convexities - such as those used to study fiscal policy transmission (Brumm et al., 2024), the effect of durable goods on the marginal propensity to consume (Kovacs et al., 2021; Beraja and Zorzi, 2024), homeownership distribution (Le Blanc et al., 2023) and labor market choices (Cruces, 2024; Bacher et al., 2024). Second, it reduces computational complexity relative to existing methods, offering considerable gains in computational time. Third, is a conceptual method rather than model-specific, providing formal guarantees for recovering the true upper envelope under general conditions. Together, these features greatly expand the class of discrete-continuous dynamic models that envelope-based methods can solve (and do so efficiently), where pre-FUES approaches failed or required restrictive assumptions.

We proceed as follows. Section 2 discusses our main results using three benchmark applications: Section 2.1 introduces FUES informally in a simple retirement choice model where agents choose savings and labor force participation; Section 2.2 demonstrates FUES when existing upper envelope methods fail, using a model where agents adjust their illiquid housing assets; Section 2.3 showcases FUES in a model with capital income tax where agents decide to hold liquid and illiquid assets via discrete choices. Section 3 formally states FUES in general form and establishes its theoretical properties. Section 4 concludes.

2 Illustrative applications

We start by discussing FUES in the context of three well-known dynamic optimization applications. After briefly introducing each problem, we discuss how their discrete choices result in non-convexity, show how FUES can retrieve the optimal solution, and compare FUES to existing applicable solution methods.

2.1 Application 1: A discrete retirement choice model

2.1.1 Model environment

We start from an application where strict monotonicity of the continuation state policy holds, which provides a clean speed and accuracy benchmark for our method. Consider the finite horizon retirement and savings choice model in Iskhakov et al. (2017) where an agent consumes, works (if they so choose), and saves from time $t = 0$ until $t = T$. At the beginning of each period, the agent starts as a worker or retiree, with the state variable denoting their beginning-of-period work status given by the discrete variable d_t. If the agent works, they earn a per-period wage y. Every period, they can choose to continue working in the next period by setting $d_{t+1} = 1$, or to permanently retire by setting $d_{t+1} = 0$. If the agent chooses to work the next period, they will incur a utility cost τ at time t. We assume all agents start as workers so $d_0 = 1$, they consume c_t, and save in financial assets a_{t+1} with $a_{t+1} \in S$ and $S := [0, \bar{a}] \subset \mathbb{R}_+$. The intertemporal budget constraint is

$$a_{t+1} = (1 + r)a_t + d_t y - c_t. \tag{1}$$

Per-period utility is given by $u(c_t) - \tau d_{t+1}$. Letting the function u be defined by $u(c) = \log(c)$, the agent's maximization problem becomes

$$V_0^{d_0}(a_0) = \max_{(c_t, d_{t+1})_{t=0}^T} \left\{ \sum_{t=0}^{T} \left[\beta^t \left(u(c_t) - \tau d_{t+1} \right) \right] \right\}, \tag{2}$$

subject to (1), $a_t \in S$ for each t, and $d_{t+1} = 0$ if $d_t = 0$ (i.e., the agent is unable to return to work after retiring).

Let $V_t^{d_t}$ denote the beginning-of-period value function. If the agent enters the period as a worker, their value function for any $t < T$ is characterized by the Bellman equation

$$V_t^1(a) = \max_{c, d_{t+1} \in \{0,1\}} \left\{ u(c) - d_{t+1}\tau + \beta V_{t+1}^{d_{t+1}}(a') \right\}, \tag{3}$$

where $a' = (1+r)a + y - c$ and $a' \in S$; at time T, their value function is $V_T^1(a) = u((1+r)a + y)$. If the agent enters the period as a retiree, the time $t < T$ value function is

$$V_t^0(a) = \max_c \left\{ u(c) + \beta V_{t+1}^0(a') \right\}, \tag{4}$$

with $a' = (1+r)a - c$; at time T, their value function is $V_T^0(a) = u((1+r)a)$.

The optimization problem for the retirees is a standard concave problem. For the workers, however, the optimization problem is not concave since they optimize jointly a discrete choice and a continuous choice. Moreover, even conditional on $d_{t+1} = 1$, next period value function V_{t+1}^1 will not be concave since the value function represents the supremum over *all future feasible combinations of discrete choices*. The non-concavity of V_{t+1}^1 produces the 'secondary kinks' described by Iskhakov et al. (2017).

To see how the choice at period t implicitly controls the future sequence of discrete choices and produces the secondary kinks, write the time t worker's value function as

$$V_t^1(a) = \max_c \max_{\mathbf{d} \in \mathbb{D}} \left\{ u(c) + \beta Q_{t+1}^{\mathbf{d}}(a') \right\}, \tag{5}$$

where $Q_{t+1}^{\mathbf{d}}$ is the $t+1$ value function conditional on a given sequence of future discrete choices $\mathbf{d}$, with $\mathbf{d} = \{d_{t+1}, d_{t+2}, \ldots, d_T\}$. In particular, letting $\bar{\boldsymbol{\tau}} = \{\tau, \beta\tau, \beta^2\tau, \ldots, \beta^{T-1}\tau\}$, we have

$$Q_{t+1}^{\mathbf{d}}(a) = \max_{(c_k)_{k=t+1}^T} \left\{ \sum_{k=t+1}^{T} \beta^{k-t-1} u(c_k) \right\} - \bar{\boldsymbol{\tau}}^{\text{tr}} \mathbf{d}. \tag{6}$$

To sum up, the value function non-concavity, even holding the choice d_{t+1} fixed, is brought on by the implicit changes in the entire future sequence of discrete choices as one controls the choice variable c in (5). In this case, the Bellman equation (3) still holds, and one can numerically implement VFI to compute a solution. The challenge arises when solving the Bellman equation using numerical methods becomes computationally burdensome. An efficient strategy in this case involves recovering the policy function by solving for points that satisfy the first order conditions (FOCs - i.e., the Euler equations) of the Bellman equation. However, since the upper envelope is not concave, the points satisfying the FOCs could be associated with any future sequence of discrete choice in Figure 1, and may not be on the upper envelope.

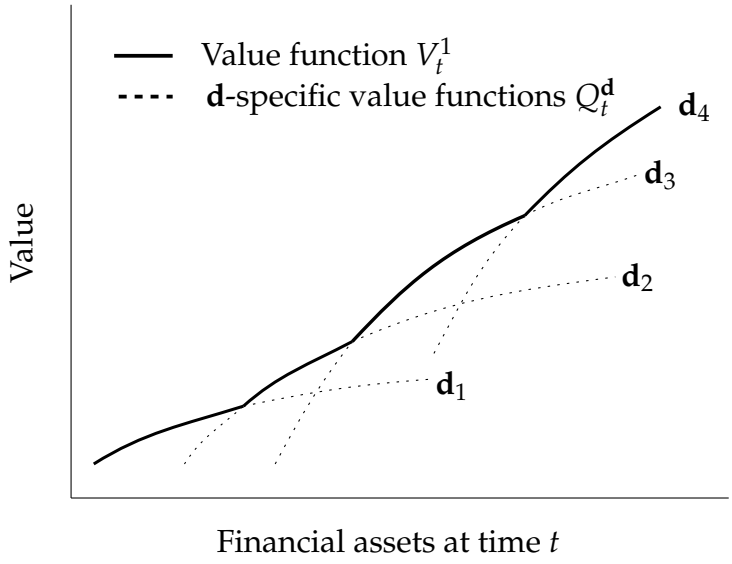


Figure 1: The time t worker value function V_t^1 is the upper envelope of concave functions, where each concave function is a value function conditional on a *sequence* of future discrete choices. The subscript 'i' on $\mathbf{d}_i$ indicates distinct sequences of future discrete choices.

2.1.2 Euler equations

In what follows, we discuss the FOCs and then showcase FUES as a way to use necessary first order information to efficiently compute V_t. If the agent chooses $d_{t+1} = 1$ (i.e., they continue as a worker in $t+1$), we can write the time t worker Euler equation as

$$u'(c_t^1) \geq \beta(1+r)u'(c_{t+1}),$$

where c_t^1 is the time t consumption policy conditional on $d_{t+1} = 1$, while c_{t+1} is the unconditional time $t+1$ consumption policy. On the other hand, if the agent chooses $d_{t+1} = 0$ (i.e., they retire), then the Euler equation is

$$u'(c_t^0) \geq \beta(1+r)u'(c_{t+1}^0),$$

where c_t^0 and c_{t+1}^0 are the consumption policies conditional on $d_{t+1} = 0$ and $d_{t+2} = 0$.

Functional Euler equations. It will now be helpful to write the Euler equation in its functional form. Let $\sigma_t^d \colon S \times \{0,1\} \to \mathbb{R}_+$ be the conditional continuation state policy function for the worker at time t if $d = 1$, and for the retiree if $d = 0$. Note that σ_t^d will depend, through its second argument, on the discrete choice (to work or not to work in $t+1$) made by the worker at time t. Unless required, for brevity, we refer to σ_t^d simply as

policy functions, with the superscript marking that it is conditional on a current discrete choice. Time t and $t+1$ policy functions will satisfy the functional Euler equation

$$u'((1+r)a + dy - \sigma_t^d(a, d_{t+1})) \\ \geq \beta(1+r)u'((1+r)\sigma_t^d(a, d_{t+1}) + d_{t+1}y - \sigma_{t+1}^{d_{t+1}}(a', d_{t+2})), \quad (7)$$

where $a' = \sigma_t^d(a, d_{t+1})$. On the choice of whether to work or not, the time t worker will choose $d_{t+1} = 1$ if and only if

$$u((1+r)a + y - \sigma_t^1(a,1)) - \tau + \beta V_{t+1}^1(\sigma_t^1(a,1)) \\ > u((1+r)a + y - \sigma_t^1(a,0)) + \beta V_{t+1}^0(\sigma_t^1(a,0)). \quad (8)$$

Since the discrete choice is itself a function of the state, define a discrete choice policy function $\mathcal{I}_t \colon S \times \{0,1\} \to \{0,1\}$. Then we have $d_{t+1} = \mathcal{I}_t(a,d)$ and $d_{t+2} = \mathcal{I}_{t+1}(a', d_{t+1})$, where $\mathcal{I}_t$ is evaluated to satisfy (8) each period conditional on the $t+1$ value function.

2.1.3 Computation using EGM and FUES

We now turn to how FUES can identify the upper envelope from a set of points that satisfy the Euler equations. Fix a time t and assume the value function V_{t+1}^d, the discrete choice function $\mathcal{I}_{t+1}$, and the optimal policy function σ_{t+1}^d for $d = 0$ and $d = 1$ are known. Let $\hat{\mathbb{X}}_t$, $\hat{\mathbb{V}}_t$ and $\hat{\mathbb{X}}_t'$ be sequences of points satisfying the following Euler equation for workers:

$$u'((1+r)\hat{x}_i + dy - \hat{x}_i') = \beta(1+r)u'((1+r)\hat{x}_i' + yd_{t+1} - \sigma_{t+1}^{d_{t+1}}(\hat{x}_i', d_{t+2})), \quad (9)$$

$$\hat{v}_i = u((1+r)\hat{x}_i + dy - \hat{x}_i') - d_{t+1}\tau + \beta V_{t+1}^{d_{t+1}}(\hat{x}_i'), \quad (10)$$

where $d_{t+2} = \mathcal{I}_{t+1}(\hat{x}_i', d_{t+1})$, $\hat{x}_i \in \hat{\mathbb{X}}_t$, $\hat{v}_i \in \hat{\mathbb{V}}_t$ and $\hat{x}_i' \in \hat{\mathbb{X}}_t'$. Such a sequence of points can be generated analytically using EGM. In particular, we have

$$(1+r)\hat{x}_i + dy - \hat{x}_i' = u'^{,-1}\left[\beta(1+r)u'((1+r)\hat{x}_i' + yd_{t+1} - \sigma_{t+1}^{d_{t+1}}(\hat{x}_i', d_{t+2}))\right]. \quad (11)$$

In the case of the EGM, $\hat{\mathbb{X}}_t$ is the endogenous grid of points, $\hat{\mathbb{X}}_t'$ is the exogenous one, and $\hat{\mathbb{V}}_t$ is the value correspondence.

Next, order the points in $\hat{\mathbb{X}}_t$, $\hat{\mathbb{V}}_t$ and $\hat{\mathbb{X}}_t'$ in ascending order of the *endogenous grid points* $\hat{\mathbb{X}}_t$. Consider the left panel of Figure 2 as a stylised plot of the endogenous grid points $\hat{\mathbb{X}}_t$ and associated continuation payoffs $\hat{\mathbb{V}}_t$ generated by EGM. Assume for the purpose

of illustration that the EGM points are associated with two overlapping future choice-specific value functions. The right panel of Figure 2 displays the policy functions (i.e., next period assets) associated with each future choice-specific value function. Pick a point $\hat{x}_i$, with $\hat{x}_i \in \hat{\mathbb{X}}_t$ such that $\hat{x}'_i$ is optimal given $\hat{x}_i$ and it lies on the upper envelope. Note if the points $\hat{x}_{i+1}$ and $\hat{x}'_{i+1}$ imply a different future sequence of discrete choices to $\hat{x}_i$ and $\hat{x}'_i$, then $\hat{x}'_{i+1}$ will experience a 'discontinuous jump' from $\hat{x}'_i$. However, for $\hat{x}'_{i+1}$ to be on the upper envelope, it must be that $\hat{x}'_{i+1}$ can only jump if it occurs after the crossing point between two value functions (for instance, the point $\hat{x}_6$). That is, $\hat{x}'_{i+1}$ can only jump if $(\hat{x}_{i+1}, \hat{v}_{i+1})$ makes a *convex* 'left turn' from the line joining $(\hat{x}_{i-1}, \hat{v}_{i-1})$ and $(\hat{x}_i, \hat{v}_i)$. On the other hand, if $(\hat{x}_{i+1}, \hat{v}_{i+1})$ makes a *concave* 'right turn', $\hat{x}'_{i+1}$ cannot jump for it to be on the upper envelope. The reason is that if $(\hat{x}_{i+1}, \hat{v}_{i+1})$ has made a right turn, for $(\hat{x}_{i+1}, \hat{v}_{i+1})$ to be on the upper envelope, it must be on the *concave value function* yielding the same future sequence of discrete choices $\mathbf{d}$ as implied by $(\hat{x}_i, \hat{v}_i)$. If a right turn is associated with a jump (e.g., point $\hat{x}_7$), then it must be on a value function associated with a suboptimal set of future discrete choices. Section 3 formally proves this argument (and provides the pseudo-code), while informally we use the intuition from Figure 2 to implement FUES as:

Box 1: FUES implementation

1. Compute $\hat{\mathbb{X}}_t$, $\hat{\mathbb{V}}_t$ and $\hat{\mathbb{X}}'_t$ using standard EGM.
2. Set a pre-determined 'jump detection' threshold $\bar{M}$.
3. Sort all sequences in order of the *endogenous* grid $\hat{\mathbb{X}}_t$.
4. Start from point $i = 2$. Compute $g_i = \frac{\hat{v}_i - \hat{v}_{i-1}}{\hat{x}_i - \hat{x}_{i-1}}$ and $g_{i+1} = \frac{\hat{v}_{i+1} - \hat{v}_i}{\hat{x}_{i+1} - \hat{x}_i}$.
5. If $|\frac{\hat{x}'_{i+1} - \hat{x}'_i}{\hat{x}_{i+1} - \hat{x}_i}| > \bar{M}$ and a right turn is made ($g_{i+1} < g_i$), then remove point $i+1$ from grids $\hat{\mathbb{X}}_t$, $\hat{\mathbb{V}}_t$ and $\hat{\mathbb{X}}'_t$. Otherwise, set $i = i + 1$.
6. If $i + 1 \leq |\hat{\mathbb{X}}_t|$, then repeat from step 5.

Note that the 'jump detection' threshold $\bar{M}$ arises naturally in economic problems since the maximum curvature of a policy function *conditional on a sequence of future discrete choices* is bounded above as marginal propensities to save and consume are also bounded above (White, 2015; Carroll and Shanker, 2025). Thus, $\bar{M}$ can simply be set to the maximum marginal propensity to consume, or it can also be analytically evaluated for each grid point (see Appendix B).

The method will thus yield a set of refined grids $\mathbb{X}_t$, $\mathbb{V}_t$ and $\mathbb{X}'_t$. To address the occasionally binding financial assets constraint $a_{t+1} \geq 0$, we follow the inverse Euler equation approach formalized by Dobrescu and Shanker (2024). In particular, we construct

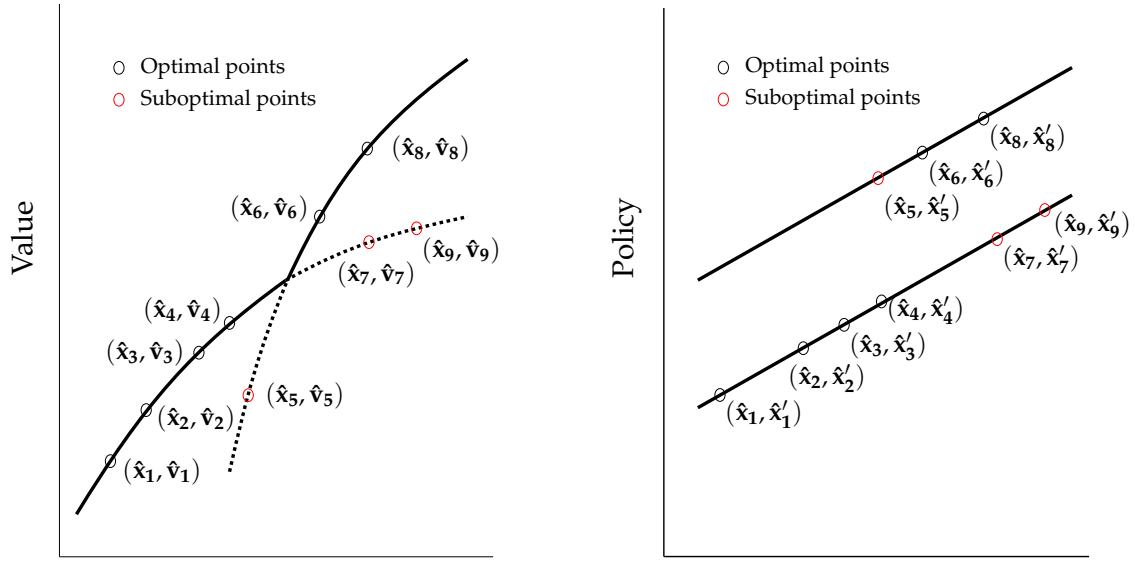


Figure 2: FUES eliminates points that cause a concave 'right turn' from an optimal point (left panel) and also cause a discontinuous jump in policy (right panel).

a constrained region of solution points as an array of consumption values c_t^1 such that $u'(c_t^1) > \beta(1+r)u'(c')$, where c' is next period consumption if $a' = 0$. The constrained region can then be appended to the unconstrained endogenous grid to create a single endogenous grid. The policy and value functions can then be interpolated over these grids to yield the time t approximated solution for the worker who chooses to continue working. The retiree value and policy functions can be calculated using standard EGM since the retiree problem is concave at each time t. Once the retiree problem is solved for time t, the discrete choice at time t of whether or not to work at time $t+1$ can be evaluated. The procedure can then be repeated at $t-1$ as per standard backward policy iteration.

Forward and backward scans and crossing points. So far, we assumed that if a point $(\hat{x}_{i+1}, \hat{v}_{i+1})$ makes a left turn from point $(\hat{x}_i, \hat{v}_i)$, then it is a sufficient and *necessary* condition for the point $(\hat{x}_{i+1}, \hat{v}_{i+1})$ to lie after a crossing point of two value functions between itself and $(\hat{x}_i, \hat{v}_i)$. However, a point $(\hat{x}_{i+1}, \hat{v}_{i+1})$ may lie after such a crossing point and yet not generate a left turn with respect to (wrt.) point x_i - see the left panel of Figure 3. We also assumed that the first point after a crossing point $(\hat{x}_{i+1}, \hat{v}_{i+1})$ must be on the optimal choice-specific value function. However, a selected point may be the first point after a crossing between choice-specific value functions and yet not be optimal - see the right panel of Figure 3 where the point $(\hat{x}_i, \hat{v}_i)$ is suboptimal but will not be removed by the basic FUES method since it is on the same sequence of future discrete choices as the

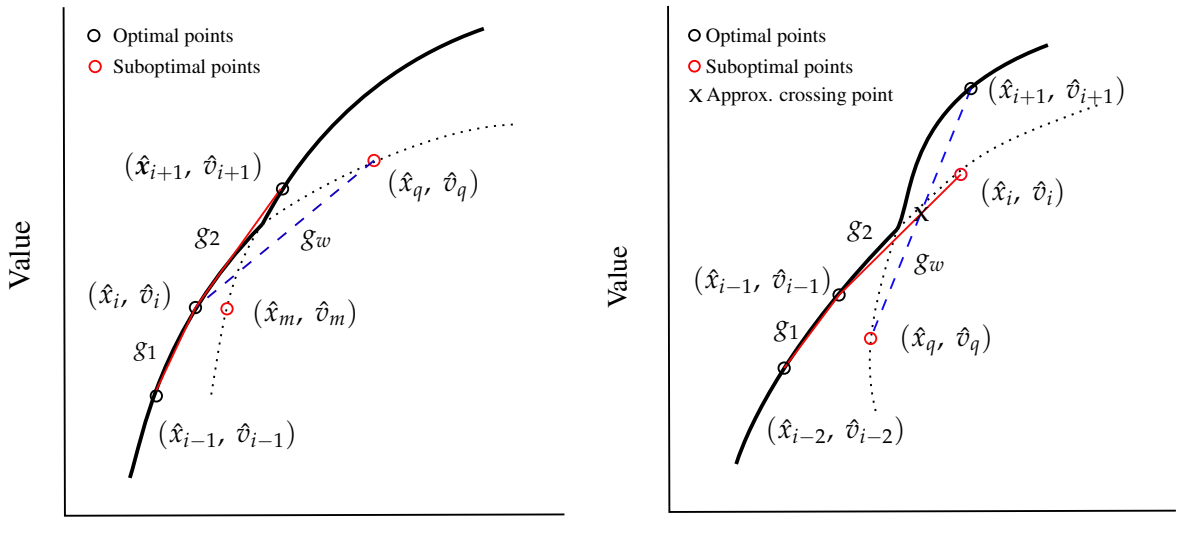


Figure 3: Forward and backward scans to improve accuracy around crossing points.

optimal point right before the crossing.

To correct the issue in the left panel of Figure 3, we can implement a forward scan before a point is eliminated. A forward scan picks a point $\hat{x}_q$ to the right of $\hat{x}_{i+1}$ that is on the same value function as $\hat{x}_i$. The point $\hat{x}_q$ can be found by checking points after $i+1$ in the endogenous grid and finding the first point whose policy function does not jump from the point $\hat{x}_i$. We can then check to see if point $\hat{x}_{i+1}$ dominates the segment joining $(\hat{x}_i, \hat{v}_i)$ to $(\hat{x}_q, \hat{v}_q)$, drawn as g_w in the left panel of Figure 3. If $\hat{x}_{i+1}$ dominates the segment, it means $\hat{x}_{i+1}$ lies after a crossing point and must be included as an optimal point.

To correct the issue in the right panel of Figure 3, we can implement a backward scan after making a left turn. A backward scan picks a point $\hat{x}_q$ to the left of $\hat{x}_i$ that is on the same value function as $\hat{x}_{i+1}$. The point $\hat{x}_q$ can be found by checking points before i in the endogenous grid and finding the first point whose policy function does not jump from the point $\hat{x}_{i+1}$. We can then check to see if point $\hat{x}_i$ is dominated by the segment joining $(\hat{x}_{i+1}, \hat{v}_{i+1})$ to $(\hat{x}_q, \hat{v}_q)$, drawn as g_w in the right panel of Figure 3. If $\hat{x}_i$ is dominated by the segment, it means $\hat{x}_i$ lies after a crossing and must not be included as an optimal point.

Finally, note that the forward and backward scan procedures can be used to attach approximations of the choice-specific value function crossing points to the endogenous grid. For instance, in the right panel of Figure 3, a crossing point can be attached at the intersection of the segment g_w with the segment g_2.

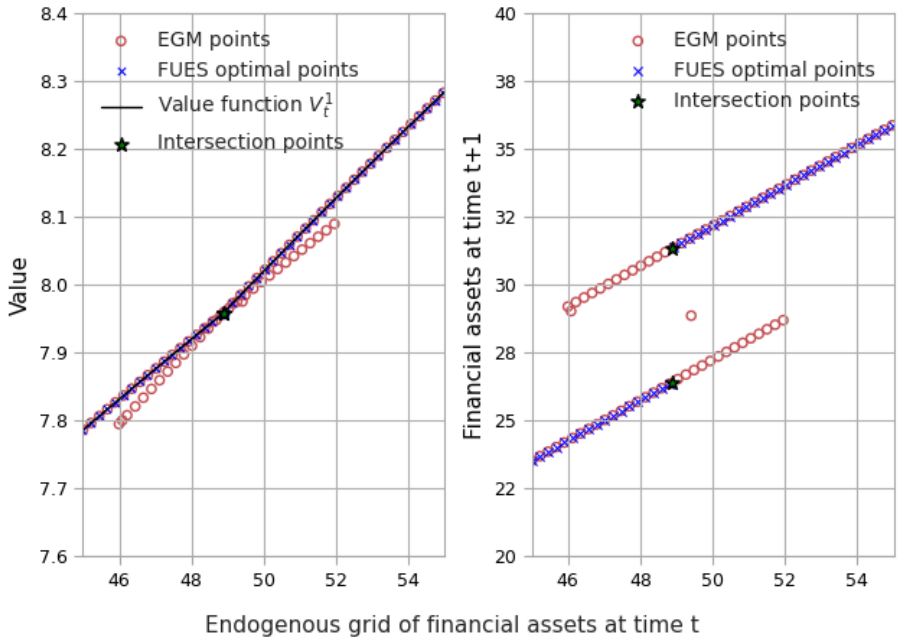


Figure 4: Value correspondence and optimal points for $t = 17$.

2.1.4 Numerical implementation

Appendix A provides a quick overview of the main upper envelope algorithms in the literature, outlines their limitations, and explains how FUES overcomes them. In what follows, we provide a numerical illustration of these arguments using the discrete retirement choice model in Iskhakov et al. (2017). Since the optimal financial assets policy function is strictly monotone, we can compare FUES with the MSS method from Iskhakov et al. (2017) using the `HARK` library (Carroll et al., 2018), the LTM method from Druedahl (2024) using the `ConsumptionSaving` package, and the RFC method from Dobrescu and Shanker (2024) using the algorithm here.[2] For FUES, we set the jump detection threshold $\bar{M}$ to 1.01, reflecting the maximum marginal propensity to consume in this model. Figure 4 shows how FUES removes suboptimal points in the value function (left panel), selects the optimal policy function (right panel), and adds the accurate intersection points.

Table 1 shows each method's upper envelope and total solution time for five different sizes of financial assets grid and utility costs associated with work, τ. Because EGM itself is efficient, the envelope step dominates total runtime, so the same ranking emerges for both metrics. Across grid sizes, FUES is between one and three orders of magnitude

[2]We use a 4th Gen. Intel Xeon processor on `Python 3.12` and just-in-time compiler `numba`. Interactive notebooks implementing all methods and replicating the results are in the online repository here.

faster than LTM, and between four and 20 times faster than the best alternative EGM-based method (MSS).[3] For example, with 1,000 asset grid points and $\tau = 1$, the envelope step completes in 0.25 milliseconds (ms) under FUES, versus 2.20 ms for MSS, 2.58 ms for RFC, and 8.1 8ms for LTM; total model solution times are 0.30 ms, 2.26 ms, 2.64 ms, and 8.23 ms, respectively. Refining the grid to 3,000 asset points increases the FUES envelope time only to 0.37ms (0.50ms total), whereas MSS, RFC and LTM rise to 5.73ms (5.87ms), 7.04ms (7.18ms) and 72.81ms (72.93ms), respectively. Thus FUES is about 15 times faster than MSS, and roughly 197 times faster than LTM on this grid. The gap widens further with a 10,000-point assets grid, where FUES finishes the envelope step in 0.82 ms (1.22 ms total) while LTM requires 798 ms (799 ms total) - a speedup factor of nearly $1,000$.

Table 1: Application 1 - Discrete retirement choice model: Speed

		FUES		MSS		LTM		RFC	
Grid	τ	U.Env.	Total	U.Env.	Total	U.Env.	Total	U.Env.	Total
1,000	0.25	0.20	0.25	0.80	0.85	8.17	8.21	2.19	2.25
	0.50	0.21	0.26	1.15	1.20	8.24	8.29	2.25	2.31
	1.00	0.25	0.30	2.20	2.26	8.18	8.23	2.58	2.64
	2.00	0.24	0.29	1.77	1.83	8.28	8.32	2.48	2.54
2,000	0.25	0.25	0.34	1.31	1.40	32.50	32.59	4.06	4.16
	0.50	0.27	0.35	1.90	2.00	32.22	32.30	4.22	4.31
	1.00	0.30	0.39	4.08	4.17	32.49	32.57	4.76	4.86
	2.00	0.31	0.39	3.33	3.42	32.31	32.39	4.67	4.77
3,000	0.25	0.30	0.43	1.81	1.94	72.88	73.00	5.95	6.09
	0.50	0.32	0.45	2.62	2.75	72.86	72.98	6.31	6.45
	1.00	0.37	0.50	5.73	5.87	72.81	72.93	7.04	7.18
	2.00	0.37	0.49	4.61	4.74	72.90	73.02	6.90	7.03
6,000	0.25	0.44	0.68	3.31	3.55	290.53	290.76	11.99	12.25
	0.50	0.48	0.72	4.80	5.05	292.82	293.05	16.04	16.29
	1.00	0.55	0.79	10.67	10.92	290.33	290.56	17.86	18.13
	2.00	0.55	0.79	8.61	8.86	291.46	291.70	17.65	17.92
10,000	0.25	0.60	0.99	5.25	5.64	800.92	801.31	24.81	25.26
	0.50	0.65	1.04	7.74	8.15	809.71	810.10	26.06	26.51
	1.00	0.82	1.22	17.14	17.54	798.49	798.89	28.90	29.35
	2.00	0.76	1.15	13.76	14.17	809.24	809.62	28.09	28.53

Notes: Parameters: $r = 0.02$, $\beta = 0.96$, $T = 50$, $y = 20$, $A_{\max} = 500$, $\bar{m} = 1.01$, $\bar{s} = 0$. **U.Env**: average upper envelope time per period. **Total**: average total solution time per period. All time in milliseconds.

[3]MSS relies on monotonicity of the continuation policy, which holds here, making it a very useful benchmark. Table 6 in Appendix D also shows results for a model with logit smoothing, yielding similar results.

The FUES speedup arises from its key numerical advantage over alternative methods: its complexity scales sub-linearly in grid size, whereas MSS scales linearly and LTM scales quadratically. Across all applications, doubling the number of grid points doubles the runtime of MSS and quadruples the upper envelope runtime of LTM. This is because FUES only computes secants from the upper envelope, whereas MSS computes linear interpolants over all possible monotone segments (including suboptimal ones) and LTM loops over all possible pairs to construct pairwise local interpolants. Importantly, the advantage that EGM has over VFI is canceled by the use of LTM for large grid sizes. Indeed, in the housing choice model (our third application), total computation time under LTM becomes slower than VFI after 4,000 grid points. With 10,000 points, total computation time under LTM is about two to three times slower than VFI. FUES, however, maintains a considerable advantage over VFI across the board.

FUES is also robust to changes in parameter values. Its timing is virtually flat in τ, whereas MSS slows down markedly as τ increases. Additionally, we also tested FUES in a version of the model where an extreme value taste shock with scale parameter $\bar{s}$ is added to the utility of workers (Iskhakov et al., 2017). FUES continues to successfully recover the upper envelope and its timing and performance remains robust. For MSS, the resulting endogenous grid in Appendix Figure 7 shows how the decreasing non-monotone parts of the endogenous grid lead to a significant number of extra short segments that need to be processed. Their presence slows down the MSS (see Appendix Table 6), with FUES now close to an order of magnitude faster than MSS, while still maintaining higher accuracy.

Finally, Table 2 reports accuracy measured as the mean log deviation against the policy implied by the Euler equation along a sample path (Judd, 1992). To evaluate how accurately FUES approximates intersection points, we also report the mean log deviation against a benchmark 'true solution' obtained using MSS with 20,000 grid points. All four methods attain essentially identical Euler equation errors, hovering around -1.63 (in $\log_{10}$) for the baseline $\tau = 1$ calibration and drifting only slightly across parameter variations. Both FUES and MSS perform equally well against the 'true solution', although LTM fares slightly worse for lower grid points. The real gain in this application is speed: FUES is always the fastest, followed by MSS, then RFC, and LTM coming in last.

2.2 Application 2: A continuous housing investment with frictions model

Consider a housing frictions model with two dimensions of savings - one for financial (liquid) assets and one for housing (illiquid) assets - where the MSS method of Iskhakov et al. (2017) and the LTM method of Druedahl and Jørgensen (2017) fail due to the non-

Table 2: Application 1 - Discrete retirement choice model: Accuracy

		FUES		MSS		LTM		RFC	
Grid	τ	E.Err	True Dev.	E.Err	True Dev.	E.Err	True Dev.	E.Err	True Dev.
1,000	0.25	-1.54	-13.04	-1.54	-13.04	-1.54	-12.22	-1.54	-13.04
	0.50	-1.56	-12.92	-1.56	-13.00	-1.56	-12.35	-1.56	-12.99
	1.00	-1.63	-12.58	-1.63	-12.87	-1.63	-12.68	-1.63	-12.79
	2.00	-1.24	-15.34	-1.25	-15.46	-1.25	-15.34	-1.25	-15.40
2,000	0.25	-1.54	-13.29	-1.54	-13.29	-1.54	-12.47	-1.54	-13.29
	0.50	-1.56	-13.28	-1.56	-13.27	-1.56	-12.62	-1.56	-13.26
	1.00	-1.63	-13.27	-1.63	-13.20	-1.63	-13.13	-1.63	-13.19
	2.00	-1.23	-15.59	-1.24	-15.54	-1.24	-15.55	-1.24	-15.53
3,000	0.25	-1.54	-13.54	-1.54	-13.54	-1.54	-12.64	-1.54	-13.54
	0.50	-1.56	-13.53	-1.56	-13.52	-1.56	-12.79	-1.56	-13.52
	1.00	-1.63	-13.52	-1.63	-13.48	-1.63	-13.32	-1.63	-13.47
	2.00	-1.23	-15.60	-1.24	-15.57	-1.24	-15.57	-1.24	-15.56
6,000	0.25	-1.54	-14.82	-1.54	-14.82	-1.54	-13.21	-1.54	-14.82
	0.50	-1.56	-14.82	-1.56	-14.82	-1.56	-13.37	-1.56	-14.81
	1.00	-1.63	-14.79	-1.63	-14.78	-1.63	-13.91	-1.63	-14.77
	2.00	-1.23	-15.60	-1.23	-15.58	-1.23	-15.59	-1.24	-15.58
10,000	0.25	-1.54	-15.02	-1.54	-15.02	-1.54	-13.97	-1.54	-15.02
	0.50	-1.56	-15.01	-1.56	-15.01	-1.56	-14.13	-1.56	-15.01
	1.00	-1.63	-14.99	-1.63	-14.99	-1.63	-14.67	-1.63	-14.98
	2.00	-1.23	-15.60	-1.23	-15.59	-1.23	-15.60	-1.23	-15.59

Notes: Parameters: $r = 0.02$, $\beta = 0.96$, $T = 50$, $y = 20$, $A_{\max} = 500$, $\bar{m} = 1.2$, $\bar{s} = 0$. **E.Err**: Euler equation error in mean $\log_{10}$ deviation. **True Dev.**: consumption deviation wrt MSS with 20,000 grid points.

monotonicity of the continuation state policy function, forcing existing approaches to rely on VFI or combining VFI with EGM (NEGM). As a result, despite being a veritable workhorse in the housing frictions literature (Kaplan and Violante, 2014; Yogo, 2016; Dobrescu et al., 2025), the conditions that Iskhakov (2015) requires for multidimensional 'analytical EGM' do not hold here. This example thus also shows how the one-dimensional EGM enabled by FUES can be applied to a multidimensional setting where analytical EGM fails but the optimal policy function is still locally invertible. The solution strategy will involve numerically obtaining all the roots of the financial assets Euler equation along a single dimension, and then evaluating the optimal non-monotone housing policy, and implicitly the optimal financial assets policy, using EGM and FUES.

2.2.1 Model environment

Let (non-negative) financial assets be denoted by a_t, and (non-negative) housing assets be denoted by H_t. Consider financial assets earn a rate of return r, and assume for simplicity that housing earns no returns. Investments can be made in and out of the stock of a_t without friction, while adjusting housing to a value H_{t+1} requires a payment of τH_{t+1}, with $\tau > 0$. In each period t, with $t = t_0, 1 \ldots, T$, the agent consumes non-housing goods c_t, and invests a total of H_{t+1} and a_{t+1} in housing and financial assets, respectively. The agent also makes a discrete choice d_t, where investment in and out of housing assets can only be made if $d_t = 1$; otherwise if $d_t = 0$ then $H_{t+1} = H_t$. Finally, in each period, the agent earns a stochastic wage y_t and we assume $(y_t)_{t=0}^T$ is a Markov process.

The following constraints will hold for each t: first, total investments and consumption cannot exceed total available wealth each period

$$(1+r)a_t + y_t + d_t H_t \geq a_{t+1} + c_t + d_t(1+\tau)H_{t+1}, \tag{12}$$

where total wealth at time t is denoted by $x_t = (1+r)a_t + y_t + d_t H_t$.

Second, in terms of payoffs, the agent lives up to time T, after which they die and value the bequest they leave behind according to a function $\theta \colon \mathbb{R}_+ \to \mathbb{R} \cup \{-\infty\}$. Per-period utility is given by a real-valued function $u \colon \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R} \cup \{-\infty\}$ as $u(c_t, H_{t+1})$, where u is a concave, jointly differentiable, increasing function of non-housing consumption c_t and end-of-period housing H_{t+1} (Yogo, 2016). Formally, the agent's dynamic optimization problem becomes

$$V_0(a_0, H_0, y_0) = \max_{(a_t, H_t, d_t)_{t=0}^{T+1}} \left\{ \sum_{t=0}^{T} \beta^t \mathbb{E} u(c_t, H_{t+1}) + \mathbb{E}\beta^{T+1}\theta((1+r)a_{T+1} + H_{T+1}) \right\}, \tag{13}$$

such that (12) holds, $H_t \geq 0$ and $a_t \geq 0$ for each t, a_0, H_0 and y_0 are given, and the expectation is taken over the wage process $(y_t)_{t=0}^T$.

$T+1$ value function is given by the bequest value $V_{T+1}(a_{T+1}, H_{T+1}) = \theta((1+r)a_{T+1} + H_{T+1})$. For each $t \leq T$, the sequential problem implies the recursive Bellman equation

$$V_t(a, H, y) = \max_{a', H', d} \left\{ u(c, H') + \beta \mathbb{E}_y V_{t+1}(a', H', y') \right\}, \tag{14}$$

where the prime notation indicates continuation state values satisfying the budget constraint (12), $c = y_t + (1+r)a + dH - a' - d(1+\tau)H'$, $H' = H$ if $d = 0$, and expectations

are now conditional on the realization y of the time t wage y_t.

2.2.2 Euler equations

The problem will feature two Euler equations, one for each state. For periods prior to the terminal one, the Euler equation for the financial assets is

$$u_1(c_t, H_{t+1}) \geq \beta(1+r)\mathbb{E}_t u_1(c_{t+1}, H_{t+2}), \tag{15}$$

where we have used the subscript '1' to refer to the first partial derivative of u, and will use the subscript '2' to refer to the second partial derivative of u. If $d_t = 1$, the Euler equation for housing is

$$(1+\tau)u_1(c_t, H_{t+1}) \geq \overbrace{\mathbb{E}_t \sum_{k=t}^{\iota-1} \beta^{k-t} u_2(c_k, H_{k+1})}^{\text{Marginal value of the housing services stream}} + \underbrace{\mathbb{E}_t \beta^{\iota-t} u_1\left(c_\iota, H_{\iota+1}\right)}_{\text{Marginal value of liquidating housing at time } \iota} . \tag{16}$$

The intuition of the Euler equation (15) for financial assets is standard. The Euler equation for housing (16), however, features a stochastic time subscript ι, defined as the next period when $d_\iota = 1$. Since the next time housing is adjusted will be stochastic, ι becomes a random stopping time. The Euler equation for housing then tells us that the shadow value (price) of investment (or withdrawal) from the stock of housing assets is given by the discounted expected value of housing *when housing is next liquidated*, along with the stream of housing services provided up to the time of liquidation.

Functional Euler equations. Since the solution sequence for the problem is recursive, there exists measurable functions σ_t^a, σ_t^H and $\mathcal{I}_t$ such that $H_{t+1} = \sigma_t^H(a_t, H_t, y_t)$, $a_{t+1} = \sigma_t^a(a_t, H_t, y_t)$ and $d_t = \mathcal{I}_t(a_t, H_t, y_t)$ for each t. Let $\sigma_t^{a,d}$ and $\sigma_t^{H,d}$ be the choice-specific policy functions conditional on the time t discrete choice $d \in \{0, 1\}$. Inserting the policy functions back into (15) - (16) yields the following functional Euler equation for housing:

$$\begin{aligned} u_1(c, H')(1+\tau) \geq\ & u_2(c, H') + \beta\mathbb{E}_y\left\{\mathcal{I}_{t+1}(a', H', y')u_1(c', H'')\right\} \\ & + \beta\mathbb{E}_y\left\{\left(1 - \mathcal{I}_{t+1}(a', H', y')\right)\left[\Theta_{t+1}(a', H', y') + u_2(c', H')\right]\right\}, \end{aligned} \tag{17}$$

where we have that

$$c' = (1+r)\sigma_t^a(a,H,y) + \sigma_t^H(a,H,y) + y' \\ - \sigma_{t+1}^a(a',H',y') - \sigma_{t+1}^H(a',H',y')(1+d_{t+1}\tau), \quad (18)$$

$$c = (1+r)a + H + y - \sigma_t^{a,1}(a,H,y) - \sigma_t^{H,1}(a,H,y)(1+\tau), \quad (19)$$

$$H'' = \sigma_{t+1}^{H,1}(a',H',y'), \quad (20)$$

$a' = \sigma_t^{a,1}(a,H,y)$, $H' = \sigma_t^{H,1}(a,H,y)$, and Θ_{t+1} is a multiplier denoting the continuation marginal value of housing if time $t+1$ housing stock is not adjusted. For a set of time $t+1$ recursive policy functions, we can compute Θ_t as a function of the states as follows:

$$\Theta_t(a,H,y) = \beta \mathbb{E}_y \left\{ \mathcal{I}_{t+1}(a',H',y') u_1(c',H'') \right\} \\ + \beta \mathbb{E}_y \left\{ (1 - \mathcal{I}_{t+1}(a',H',y')) \left[\Theta_{t+1}(a',H',y') + u_2(c',H') \right] \right\}. \quad (21)$$

The functional Euler equation for the financial assets then becomes

$$u_1(c,H') \geq \beta(1+r)\mathbb{E}_y u_1(c',H''), \quad (22)$$

with c, c' H' defined analogously to (18) and (19), and where if the time t discrete choice is not to adjust, then $d = 0$.

2.2.3 Computation using EGM and FUES

Suppose we know V_{t+1}, $\{\sigma_{t+1}^{a,d}, \sigma_{t+1}^{H,d}\}_{d\in\{0,1\}}$, $\mathcal{I}_{t+1}$ and Θ_{t+1}. We can first apply standard EGM with FUES to evaluate $\sigma_t^{a,0}$ for housing non-adjusters since only one Euler equation - i.e., equation (22) - will hold. For each possible time t housing state H_t in the housing grid and exogenous state y_t, we can approximate $\sigma_t^{a,0}(\cdot, H_t, y_t)$ by first setting an exogenous grid of a' values (holding $H' = H_t$ fixed since no adjustment takes place), and then creating an endogenous grid of time t financial assets using (22) and (12). FUES can then be applied as in Section 2.1.3 to eliminate suboptimal points, allowing us to obtain an approximation of the policy function $\sigma_t^{a,0}$.[4]

Let us now calculate the housing policy functions for housing adjusters. First, note we can evaluate consumption today c, for a given value of H' and a', as

$$c = u_1^{-1}((1+r)\beta\mathbb{E}_y u_1(c',H''),H'), \quad (23)$$

[4]To address occasionally binding constraints, we follow the approach in Dobrescu and Shanker (2024).

where u_1^{-1} is the analytical inverse of u_1 in its first argument. We evaluate the policy functions for adjusters as follows:

Box 2: FUES and EGM for the adjuster's policy function

1. Fix y, fix an exogenous grid $\tilde{\mathbb{H}}'$ over H', and initialise empty arrays for the housing policy ($\hat{\mathbb{H}}'$), value correspondence ($\hat{\mathbb{V}}$), financial assets policy function ($\hat{\mathbb{A}}'$), and endogenous wealth grid ($\hat{\mathbb{X}}$).
2. For each $\hat{h}_i \in \tilde{\mathbb{H}}'$:
   (i) Evaluate the $P+1$ multiple roots to (17) in terms of a', with c evaluated by (23) and H' fixed as $\hat{h}_i$; collect the roots in a tuple $(\hat{a}'^{,0}, \ldots, \hat{a}'^{,p}, \ldots \hat{a}'^{,P})$.
   (ii) For each root in $(\hat{a}'^{,0}, \ldots, \hat{a}'^{,p}, \ldots, \hat{a}'^{,P})$, evaluate the endogenous grid points of wealth $(\hat{x}^0, \ldots, \hat{x}^p, \ldots, \hat{x}^P)$ using the budget constraint[a]

$$\hat{x}^p = \hat{a}'^{,p} + (1+\tau)\hat{h}_i + c,$$

   and evaluate the current period values $(\hat{v}'^{,0}, \ldots, \hat{v}'^{,p}, \ldots \hat{v}'^{,P})$ as

$$\hat{v}'^{,p} = u(c, \hat{h}_i) + \beta \mathbb{E}_y V_{t+1}(\hat{a}'^{,p}, \hat{h}_i, y').$$

   (iii) Append the $P+1$ multiple roots $(\hat{a}'^{,0}, \ldots, \hat{a}'^{,p}, \ldots \hat{a}'^{,P})$ to $\hat{\mathbb{A}}'$, the time t values to $\hat{\mathbb{V}}$, the endogenous grid points to $\hat{\mathbb{X}}$, and $P+1$ copies of $\hat{h}_i$ to $\hat{\mathbb{H}}'$.
3. Apply FUES (Box 1, Section 2.1.3) to the grids $\hat{\mathbb{V}}$, $\hat{\mathbb{X}}$ and $\hat{\mathbb{H}}$ to recover the refined endogenous grid points $\mathbb{X}$, value grid $\mathbb{V}$, and policy grids $\mathbb{A}'$ and $\mathbb{H}'$.

[a] To save computation time, the endogenous grid for adjusters is defined in terms of total wealth $x = (1+r)a + y + H$ since once housing is liquidated, only x will affect the agent's decision.

We can apply the above steps to each y in the exogenous shock grid. With $\sigma_t^{H,1}$ and $\sigma_t^{a,1}$ approximated on a uniform grid, we can then construct σ_t^H and σ_t^a in the standard way, by comparing the value function for adjusters and non-adjusters at each point on the uniform grid of current period states.

Non-monotonicity of the policy functions. Before we turn to the numerical example, let us discuss how the non-monotonicity of the endogenous grid points arises for the housing-adjusting policy functions. The EGM above generates an exogenous grid $\hat{\mathbb{H}}'$ and

an endogenous grid $\hat{\mathbb{X}}$ that satisfy

$$u_1(\hat{c}, \hat{H}')(1+\tau) = \beta \mathbb{E}_y \left[\mathcal{I}_{t+1}(\hat{\psi}_t(\hat{x}, \hat{H}'), \hat{H}', y') u_1(c', H'') \right] + u_2(c, H') \\ + \beta \mathbb{E}_y \left\{ (1 - \mathcal{I}_{t+1}(\hat{\psi}_t(\hat{x}, \hat{H}'), \hat{H}', y')) \left[\Theta_{t+1}(\hat{\psi}_t(\hat{x}, \hat{H}'), \hat{H}', y') + u_2(c', H') \right] \right\}, \quad (24)$$

where

$$\hat{c} = \hat{x} - \hat{\psi}_t(\hat{x}, \hat{H}') - (1+\tau)\hat{H}', \quad (25)$$

and c' and H'' are defined as in (18) - (20). The function $\hat{\psi}_t$ gives the value of a' that satisfies the financial assets Euler equation (22), for a given value of $\hat{x}$ and $\hat{H}'$. To see the source of the non-monotonicity of H' in terms of the $\hat{x}$, first note how holding $\hat{\psi}_t$ fixed, the term on the LHS of (24) falls as wealth $\hat{x}$ increases, resulting in a higher level of housing investment. However, allowing $\hat{\psi}_t$ to also adjust alongside $\hat{x}$ implies that $\hat{\psi}_t$ may experience a discontinuous jump up, causing a discontinuous rise in the marginal utility of consumption today - i.e., the RHS of (22) jumps up. As a result, the LHS of (24) jumps up, leading to a discontinuous fall in housing investment - i.e., if the agent reaches in the future a certain wealth level and upgrades their housing stock, they will discontinuously decrease their current housing and non-housing consumption as the intertemporal substitution effect leads them to discontinuously increase financial savings.

2.2.4 Numerical implementation

To parameterize the model, we use the following separable specification for non-housing consumption and housing:

$$u(c, H) = \alpha \frac{c^{\gamma_C - 1} - 1}{\gamma_C - 1} + (1-\alpha) \frac{\kappa H^{\gamma_H - 1} - 1}{\gamma_H - 1}, \quad (26)$$

and define bequests as $\theta(c) = \bar{\theta} u(c, 0)$. For simplicity, assume agents start working at $t_0 = 20$ and while working, they earn salary $y_t = \mathrm{y}_t(\tilde{z})$, where y_t is the wage function for females in Dobrescu et al. (2025) and $\tilde{z}$ is a shock that follows an AR1 process. We also assume agents retire at age of 60 and live until they are $T = 70$ years old.

The map of endogenous grid points to the exogenous grid $\mathbb{\hat{H}}'$ for a 59-year-old is shown in Figure 5. (Appendix Figure 8 further compares policy functions computed via NEGM with those computed via FUES and EGM.) First, note that there are multiple endogenous grid points for each value of the exogenous grid. Second, the endogenous grid has no discernible monotone segments of consecutive covering intervals that can be identified by the MSS and LTM. Third, the optimal endogenous grid 'jumps down', implying the

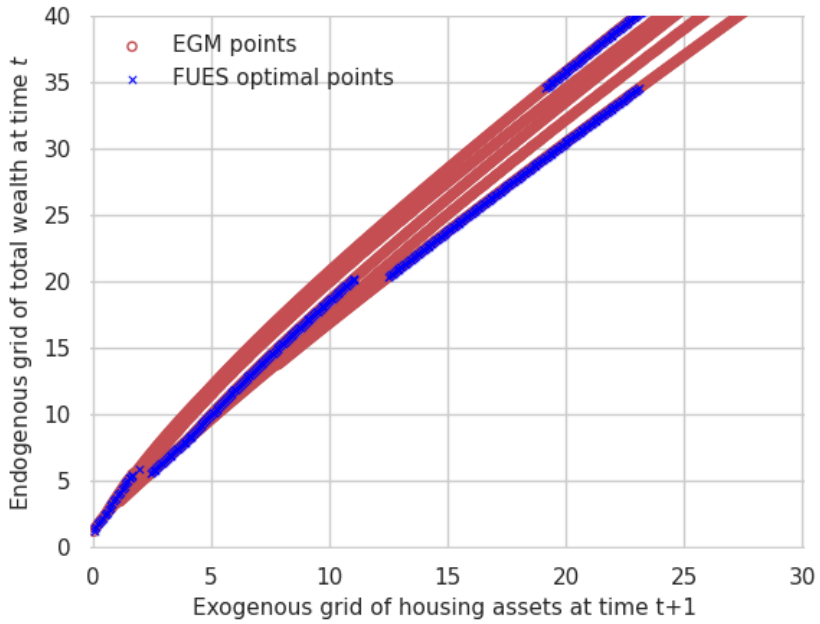


Figure 5: EGM points and non-monotone optimal policy for total wealth plotted on a uniform exogenous grid.

policy function is not monotone.

As a result, neither MSS nor LTM can be applied: MSS fails since the continuation value no longer enjoys the single crossing property required by Iskhakov et al. (2017) Theorem 4. As shown in Figure 5, a non-monotone housing policy leads to exogenous grid points mapping to multiple endogenous grid points - i.e., the optimal set of endogenous points 'bends backwards' on the exogenous grid. MSS can thus no longer scan along the exogenous grid to detect monotone segments by detecting where the endogenous grid decreases. LTM also fails since each point in the decision state will have multiple covering intervals of points in the exogenous grid, and each interval may not be formed from pairs of points over a common segment. In contrast, FUES continues to work, implementing EGM to get the upper envelope of the points recovered from inverting the Euler equation on the exogenous grid without requiring any structure. We thus use Brent's method with bracketing and select a bracketing interval of 0.01 to recover the multiple roots of the Euler equation (17). To keep things tractable, we benchmark EGM with FUES against NEGM - the main 'competitor' method in this case that has already been shown to offer significant accuracy and speed benefits (Druedahl, 2021). NEGM first approximates the non-adjuster policy function evaluated using EGM, then evaluates the upper envelope,

and subsequently solves the adjuster problem using VFI.[5]

Table 3: Application 2 - Continuous housing choice model: Speed & accuracy

		EGM-FUES					NEGM				
		Time		E.Err			Time		E.Err		
N_G	τ	N-adj	Adj	Avg	N-adj	Adj	N-adj	Adj	Avg	N-adj	Adj
250	0.05	238	5	-4.59	-4.61	-4.26	239	10	-4.40	-4.49	-2.38
	0.07	237	5	-4.60	-4.61	-4.25	238	10	-4.46	-4.52	-2.42
	0.12	238	5	-4.84	-4.85	-4.30	238	10	-4.82	-4.86	-2.42
500	0.05	1539	16	-5.74	-5.77	-4.90	1516	21	-5.46	-5.57	-2.74
	0.07	1513	15	-5.91	-5.94	-4.91	1516	21	-5.70	-5.79	-2.75
	0.12	1534	15	-6.21	-6.23	-4.94	1508	21	-6.05	-6.11	-2.76
750	0.05	4687	29	-6.31	-6.36	-5.25	4676	33	-5.98	-6.10	-2.91
	0.07	4667	29	-6.50	-6.54	-5.34	4661	33	-6.21	-6.31	-2.87
	0.12	4716	28	-6.78	-6.80	-5.30	4674	32	-6.63	-6.69	-2.96
1,000	0.05	10473	50	-6.69	-6.73	-5.49	10304	44	-6.32	-6.46	-3.05
	0.07	10488	52	-6.82	-6.86	-5.61	10497	44	-6.51	-6.61	-3.04
	0.12	10465	50	-7.09	-7.12	-5.54	10473	45	-6.86	-6.92	-3.12

Notes: Parameters: $\beta = 0.94$, $\gamma_C = 3.5$, $\gamma_H = 1.5$, $\alpha = 0.70$, $\delta = 0.00$, $r = 0.04$, $r_H = 0.00$, $\rho_w = 0.82$, $\sigma_w = 0.11$, $N_{sim} = 10,000$. Housing H and asset a grid size is N_G. **Time**: average total solution time per period. **E.Err**: Euler equation error in mean $\log_{10}$ deviation. All time in milliseconds.

The numerical performance of EGM with FUES is shown in Table 3 for different sizes of the housing and financial assets grid and various transaction costs τ. Since NEGM is not an upper envelope method whose timing can be directly compared to FUES, Table 3 compares total solution times for the non-adjuster (N-adj) and adjuster (Adj) problem under NEGM and under EGM with FUES. Despite the root-finding operations performed by FUES, this method is on average 50% faster than NEGM in solving the adjuster's problem. FUES also yields roughly two to 2.5 orders of magnitude better accuracy for the adjuster problem over NEGM (e.g., -4.94 vs. -2.76 Euler error in $\log_{10}$ at grid size 500).[6] The performance of FUES remains consistent across grid sizes and τ values. As expected, FUES improves only slightly the non-adjuster Euler error since NEGM also applies EGM to solve the non-adjuster's problem: both methods achieve similar accuracy around -6

[5]The NEGM implementation uses FUES to compute the non-adjuster upper envelope, allowing a direct comparison of NEGM's numerical optimization with EGM. Since the non-adjuster policy function is monotone given H', MSS and LTM can also be applied to the nested non-adjuster problem. Using them neither improves accuracy of the non-adjuster solution, nor brings any speed gains compared to FUES (see here).

[6]We compute the adjuster Euler error using (17). To approximate Θ_{t+1} in the RHS of (17), we recursively evaluate (21) on a finer grid with four times more points in each dimension than the solution grid.

to -7. Because adjusters are a relatively small fraction (about 1-2%) of agents in the simulated time series, the improvement in the combined Euler error across all agents is more modest. Nevertheless, the accuracy gains for adjusters demonstrate that FUES successfully enables EGM-level precision in models where policy functions are non-monotone.

2.3 Application 3: A discrete housing choice model

We now turn to the housing choice model of Fella (2014), in which owner-occupied housing is selected from a discrete grid. We extend the within-period problem in two ways. First, we introduce an explicit renting alternative, increasing the number of discrete options and bringing the decision structure closer to multi-stage discrete-continuous problems in larger applied models. We can thus assess if improvements in the upper envelope step translate into *end-to-end* computational gains once other components of the solution routine (e.g., evaluating multiple discrete alternatives, computing conditional expectations, etc.) account for a larger share of total computation time. Second, we introduce a non-smooth income tax applied to capital returns, modeled as a piecewise-linear schedule with kinks and notches. Such threshold structures are ubiquitous in tax-and-transfer systems (Saez, 2010; Chetty et al., 2011; Kleven and Waseem, 2013; Kleven, 2016; Best and Kleven, 2018; Jones and Li, 2018) and provide a transparent mechanism for generating non-strictly monotone savings policies. With income uncertainty and discrete housing choice, grid refinement quickly renders VFI impractical, while non-monotonicity rules out MSS and LTM implementations - making an efficient envelope operator essential.

2.3.1 Model environment

Consider an agent who draws a sequence of bounded, stationary Markov wage shocks $(y_t)_{t=0}^{T}$ taking values in a finite discrete space $Y \subset \mathbb{R}_+$. The agent enters each period with housing stock H_t and financial assets a_t, and decides if they will be a renter ($H_{t+1} = 0$) or homeowner ($H_{t+1} > 0$). They then decide how much to consume of non-durable goods c_t, how much to save in financial assets a_{t+1}, how much to save in housing assets H_{t+1} if they are homeowners, and how much housing services S_t to buy if they rent. Similar to Application 2, housing assets serve a dual role: they are a form of investment good but also provide durable consumption services. Housing can only be purchased in discrete amounts on a finite grid $\mathbb{H} \subset \mathbb{R}_{++}$, and adjusting housing each period incurs a fixed transaction cost τH_{t+1}, with $\tau \in [0, 1)$. We assume no borrowing and so, $a_t \geq 0$.

Financial assets returns are subject to a piecewise-linear tax. Let $T(a)$ denote total tax

liability on assets a, defined over K brackets as

$$T(a) = B_k + \tau_k(a - a_k^0), \quad \text{for } a \in [a_k^0, a_k^1],\ k = 1, \ldots, K, \tag{27}$$

where a_k^0, a_k^1 are the bracket thresholds, τ_k is the marginal tax rate on assets in bracket k, and B_k is a fixed component ensuring either continuity or an upward or downward discrete jump at the bracket boundary.[7] Since the capital rate of return r is fixed, the marginal rate τ_k relates to an equivalent income-based rate via τ_k/r, so the tax can be interpreted as levied on capital income ra.[8]

The budget constraint for a homeowner entering period t with housing stock H_t is

$$a_{t+1} + c_t \;=\; (1+r)a_t - T(a_t) + y_t + d_t^{\text{adj}} H_t - d_t^{\text{adj}}(1+\tau)H_{t+1} \;:=\; w_t^{\text{own}}, \tag{28}$$

where $d_t^{\text{adj}} := \mathbb{1}_{\{H_{t+1} \neq H_t\}}$ indicates a housing stock adjustment and w_t^{own} denotes the homeowner's total wealth net of housing expenses. If the agent instead rents in period t, the budget constraint is

$$a_{t+1} + c_t \;=\; (1+r)a_t - T(a_t) + y_t + H_t - P^r S_t \;:=\; w_t^{\text{rent}}, \tag{29}$$

where S_t denotes housing services bought at price P^r, and w_t^{rent} is the renter's total wealth net of housing expenses.

People can 'consume' housing services by either renting or owning a house and so, following Fella (2014), per-period utility is

$$u(c_t, h_t) \;=\; \alpha \log c_t + (1-\alpha)\log\big[\kappa(h_t + \iota)\big], \qquad \kappa > 0,\ \iota > 0 \tag{30}$$

where $h_t = H_{t+1}$ for homeowners and $h_t = S_t$ for renters. Turning to the recursive problem, let V_{t+1} denote the continuation value at the end of time t. For any beginning-of-period state (a, y, H), we have

$$V_t(a, y, H) \;=\; \mathbb{E}_t \left\{ \sum_{j=t}^{T} \beta^{j-t}\, u\big(c_j, h_j\big) + \beta^{T+1-t}\theta((1+r)a_{T+1} + H_{T+1}) \right\}, \tag{31}$$

[7] Allowing level shifts at thresholds accommodates *notches* (discrete changes in net tax liability) that arise in many tax-and-transfer systems - e.g., eligibility for fixed credits/rebates, asset-tested benefits, exemptions, bracketed surcharges. A negative (positive) jump corresponds to a one-off reduction (increase) in tax liability upon crossing the threshold.

[8] This non-smooth capital income tax example provides a simple benchmark, but it is worth noting that FUES readily accommodates similar - and often more intricate - non-monotonicities such as those arising under non-smooth labor income taxation or from interactions across multiple asset-tax provisions.

where $c_j = \sigma_j^c(a_j, y_j, H_j)$ and $h_j = \sigma_j^h(a_j, y_j, H_j)$ are the beginning-of-period policy functions for consumption and housing services at time j, and θ is the bequest function. Each period can be broken down into multiple stages, proceeding backwards from the end of a period as follows: first, given housing stock H_{t+1}, the homeowner solves the consumption stage as

$$\hat{V}_t^{\text{own}}(H_{t+1}, y_t, w_t^{\text{own}}) \;=\; \max_{c_t \le w_t^{\text{own}}} \Big\{ u(c_t, H_{t+1}) + \beta\, \mathbb{E}_{y_t} V_{t+1}(a_{t+1}, y_{t+1}, H_{t+1}) \Big\}, \tag{32}$$

subject to $a_{t+1} + c_t = w_t^{\text{own}}$, where $\hat{V}_t^{\text{own}}$ is the homeowner's current value function. Second, given a choice of rented services S_t, the renter solves the consumption stage as

$$\hat{V}_t^{\text{rent}}(S_t, y_t, w_t^{\text{rent}}) \;=\; \max_{c_t \le w_t^{\text{rent}}} \Big\{ u(c_t, S_t) + \beta\, \mathbb{E}_{y_t} V_{t+1}(a_{t+1}, y_{t+1}, 0) \Big\}, \tag{33}$$

subject to $a_{t+1} + c_t = w_t^{\text{rent}}$. Third, the homeowner moves to the housing choice stage, choosing H_{t+1} to maximize

$$V_t^{\text{own}}(a_t, y_t, H_t) \;=\; \max_{H_{t+1}} \hat{V}_t^{\text{own}}(H_{t+1}, y_t, w_t^{\text{own}}), \tag{34}$$

subject to (28) and $w_t^{\text{own}} \ge 0$. Analogously, the renter chooses S_t to maximize

$$V_t^{\text{rent}}(a_t, y_t, H_t) \;=\; \max_{S_t} \hat{V}_t^{\text{rent}}(S_t, y_t, w_t^{\text{rent}}), \tag{35}$$

subject to (29) and $w_t^{\text{rent}} \ge 0$. Finally, the decision to own or rent is made,

$$V_t(a_t, y_t, H_t) \;=\; \max\big\{ V_t^{\text{own}}(a_t, y_t, H_t),\; V_t^{\text{rent}}\big(a_t, y_t, H_t\big) \big\}. \tag{36}$$

2.3.2 Euler equations

To derive the necessary Euler equation for the consumption choice stages (32)–(33), let σ_{t+1}^a and σ_{t+1}^H denote time $t+1$ policy functions mapping financial assets and housing states (a_{t+1}, H_{t+1}) to choices (a_{t+2}, H_{t+2}). For an interior solution $a' \in (a_k^0, a_k^1)$ within bracket k, the Euler equation holds with equality,

$$u'(c_t) = \mathbb{E}_{y_t}(1 + r - \tau_k)\beta u'(c_{t+1}), \quad a' \in (a_k^0, a_k^1), \tag{37}$$

where $c_t = (1+r)a_t - T(a_t) + y_t - \Phi(H_t, H_{t+1}) - a'$, $c_{t+1} = (1+r)a' - T(a') + y_{t+1} - \Phi(H_{t+1}, H_{t+2}) - \sigma_{t+1}^a(a', H_{t+1})$, and $\Phi(H_t, H_{t+1}) = d_t^{\text{adj}}\big((1+\tau)H_{t+1} - H_t\big)$. At the bound-

ary between brackets k and $k+1$ (i.e., $a_k^1 = a_{k+1}^0$), the Euler condition holds as a one-sided inequality,

$$\begin{aligned} u'(c_t) &\leq \mathbb{E}_{y_t}(1+r-\tau_k)\beta\, u'(c_{t+1}), \qquad a' = a_k^1, \\ u'(c_t) &\geq \mathbb{E}_{y_t}(1+r-\tau_{k+1})\beta\, u'(c_{t+1}), \qquad a' = a_{k+1}^0. \end{aligned} \tag{38}$$

For a kink without a jump, (i) when $\tau_{k+1} > \tau_k$, the kink is convex and the value function remains concave, while (ii) when $\tau_{k+1} < \tau_k$, the kink is non-convex and introduces a convex segment in the value function. For a kink with a jump, such jumps similarly generate non-concavities of the continuation value.

2.3.3 Computation using EGM and FUES

To construct the endogenous grid, we follow Dobrescu and Shanker (2024) that shows that the endogenous grid consists of all points satisfying FOCs (with equality or inequality).

Box 3: EGM with tax kinks for the homeowner problem

1. Fix H' and y, fix a uniform exogenous grid $\tilde{\mathbb{A}}'$ over a' values, and initialize empty arrays for the value correspondence ($\hat{\mathbb{V}}$), consumption policy ($\hat{\mathbb{C}}$), financial assets policy ($\hat{\mathbb{A}}'$), and endogenous wealth grid ($\hat{\mathbb{M}}$).
2. For each $\tilde{a}_i' \in \tilde{\mathbb{A}}'$, invert (37) to obtain $\hat{c}_i$ and evaluate $\hat{m}_i = \hat{c}_i + \tilde{a}_i'$ and $\hat{v}_i = u(\hat{c}_i, H') + \beta \mathbb{E}_y V_{t+1}(\tilde{a}_i', y', H')$. Append $\hat{c}_i$ to $\hat{\mathbb{C}}$, $\tilde{a}_i'$ to $\hat{\mathbb{A}}'$, $\hat{m}_i$ to $\hat{\mathbb{M}}$, and $\hat{v}_i$ to $\hat{\mathbb{V}}$.
3. At each $a^* = a_k^1 (= a_{k+1}^0)$: for a range $c \in [c^*(1-\eta_c), c^*(1+\eta_c)]$ around c^*, retain c if (38) holds and append $(c + a^*,\ u(c, H') + \beta \mathbb{E}_y V_{t+1}(a^*, y', H'),\ a^*,\ c)$ to grids.[a]
4. Let $\hat{z}_i$ denote the tax-bracket index for $\hat{a}_i$, with $\hat{z}_i \in \hat{\mathbb{Z}}$ for a sequence $\hat{\mathbb{Z}}$. To apply FUES, replace jump detection by $\left(\left| \frac{\hat{a}_{i+1}' - \hat{a}_i'}{\hat{m}_{i+1} - \hat{m}_i} \right| > \bar{M} \ \vee\ \hat{z}_{i+1} \neq \hat{z}_i \right)$.

[a] Grid-spacing points in $[c^*(1-\eta_c), c^*(1+\eta_c)]$ are matched to the spacing of points in the exogenous grid and η_c is chosen to be equal to the maximum tax bracket size.

Non-strict monotonicity of the policy functions. We apply FUES directly to the raw unrefined EGM grid, and plot (i) the associated value and policy correspondences in Figure 6, and (ii) the tax brackets in Appendix Figure 9 (see Section 2.3.4 for numerical details). Figure 6 clearly shows why standard upper envelope routines are not directly applicable in the presence of tax kinks and notches. The right panel plots the correspondence of unrefined a' values over the endogenous grid of wealth. At tax-bracket boundaries, flat segments form where tax bracket constraints bind. Over these segments, multiple wealth

levels map to the same optimal savings choice. This violates the strict monotonicity required by MSS: when the optimal policy is constant, it can form decreasing segments of candidate solution points on the exogenous grid. Detecting drops in the endogenous grid value can then no longer separate distinct segments. Even abstracting from identifying constrained solutions, MSS will still fail to distinguish segments belonging to separate tax brackets because the endogenous grid can jump upwards (for instance, when the marginal tax rate falls upon entering a new bracket). LTM encounters a similar difficulty, as it relies on policy invertibility to construct covering intervals (see Appendix A).

To circumvent these issues, one could instead run EGM separately within each tax bracket k, and 'stitch' together bracket-specific interpolants along with additional interpolants for the constrained segments generated at bracket boundaries (Druedahl and Jørgensen, 2017). This application-specific strategy quickly becomes computationally burdensome (Dobrescu and Shanker, 2024) as each bracket can produce multiple segments with distinct interpolation domains, all of which must be stored, evaluated and compared. Moreover, the direction of these policy and value correspondence jumps across brackets is generally ambiguous and may change across iterations, depending on the relative strength of income and substitution effects. In contrast, the necessary conditions underlying Box 3 are sufficient for FUES to operate directly on the raw EGM grid and recover the upper envelope in a single well-defined scan, without requiring strict monotonicity or global invertibility of the optimal policy.

2.3.4 Numerical implementation

We present numerical results in two parts. First, we benchmark FUES against MSS, LTM, and VFI using the Fella (2014) model with renting but *without* the tax extension. This comparison isolates the performance gains from FUES in this richer framework. We then introduce the piecewise-linear tax schedule described above and fully flesh out the FUES implementation in settings where policy function monotonicity breaks down.

Model without capital income tax. Assume that agents consume all their wealth at the end of their life and compute the problem backwards from T by solving for the homeowner consumption choice in (32) and the renter consumption choice in (33) separately using EGM. Our computation parameters are similar to those in Fella (2014) - see Table 4, with two exceptions. First, we extend the numerical experiments in Fella (2014) by considering grids with up to 10,000 points to assess the scale of the possible accuracy gains. Second, while FUES can easily be applied to infinite horizon models as that in Fella (2014),

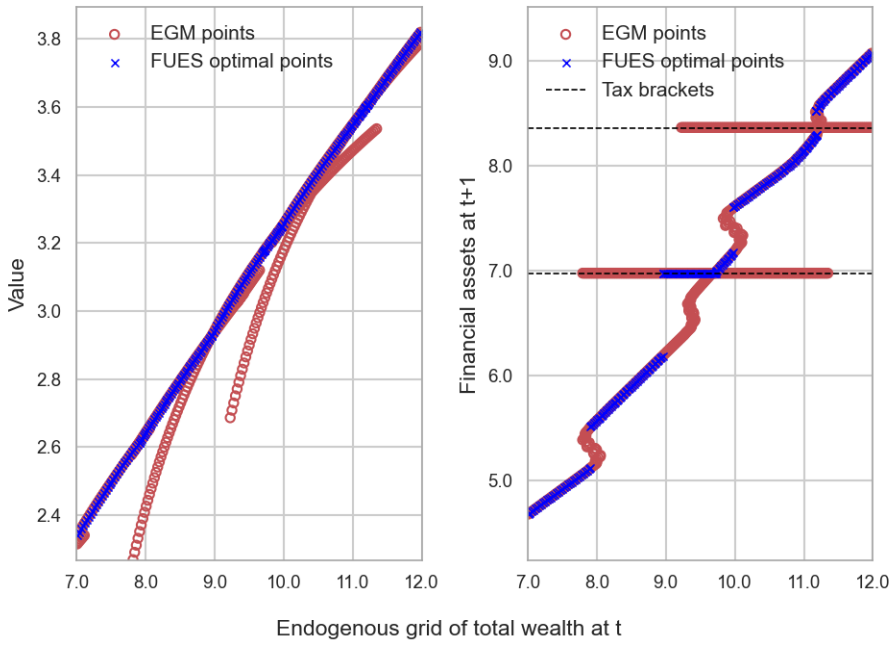


Figure 6: Value and policy correspondences with tax notches and kinks.

here we focus on a finite horizon setting for simplicity (i.e., to avoid potential complications from non-convergence of 'infinitely many jumps'). The results from a one-to-one implementation of Fella (2014) are in Appendix Table 7.

Table 4 reports computational speed and Euler equation errors across grid sizes and methods. At coarser grids, the accuracy advantage of EGM-based methods over VFI is similar to that reported by Fella (2014): with 1,000 asset grid points, Euler errors fall by roughly 2.5 orders of magnitude. At finer grids, however, these gains compound: moving from 1,000 to 10,000 points delivers an additional 1.9–2.0 orders of magnitude accuracy improvement for EGM-based methods, compared to only one order of magnitude for VFI. This widens the accuracy gap from 2.5 to approximately 3.5 orders of magnitude - a substantial advantage for applications requiring high precision.

How do these accuracy gains weigh against runtime? Table 4 reports (i) per-period upper envelope time (U.Env), (ii) runtime in the homeowner consumption stage (Cons) - i.e., the per-period cost of solving the consumption problem in (32), and (iii) per-period total solution time (Total). As expected, the algorithms exhibit distinct scaling behavior: upper envelope time under FUES scales less than linearly, MSS scales more than linearly, and LTM scales approximately quadratically. Crucially, due to the larger number of secondary jumps in this framework, the performance gap between FUES and competing methods is

Table 4: Application 3 - Discrete housing choice model without taxes: Speed & accuracy

Grid		FUES				MSS				LTM				VFI		
W	H	U.Env	Cons.	Total	E.Err	U.Env	Cons.	Total	E.Err	U.Env	Cons.	Total	E.Err	Cons.	Total	E.Err
500	5	0.04	0.05	0.38	-5.42	0.38	0.20	0.69	-5.43	1.01	0.52	1.26	-5.44	3.33	6.87	-3.22
	7	0.06	0.07	0.43	-5.41	0.55	0.28	0.90	-5.42	1.44	0.74	1.72	-5.42	4.68	9.62	-3.22
	10	0.08	0.10	0.51	-5.41	0.79	0.40	1.22	-5.41	2.05	1.06	2.39	-5.42	6.48	13.27	-3.21
1,000	5	0.05	0.07	0.44	-6.00	0.80	0.38	1.18	-6.01	3.97	2.02	4.29	-6.01	6.67	13.64	-3.52
	7	0.08	0.10	0.55	-6.00	1.24	0.58	1.72	-6.00	5.67	2.88	6.05	-6.01	9.61	19.49	-3.52
	10	0.12	0.15	0.74	-5.99	1.83	0.83	2.49	-6.00	8.07	4.11	8.65	-6.01	13.06	26.64	-3.52
2,000	5	0.08	0.11	0.60	-6.60	1.72	0.79	2.26	-6.60	16.34	8.23	16.78	-6.60	13.58	27.56	-3.81
	7	0.11	0.16	0.83	-6.60	2.73	1.22	3.53	-6.59	23.84	12.01	24.53	-6.61	19.52	39.67	-3.81
	10	0.16	0.24	1.22	-6.60	4.43	1.91	5.62	-6.59	32.54	16.41	33.51	-6.60	26.91	54.46	-3.81
4,000	5	0.13	0.20	1.03	-7.18	3.91	1.74	4.91	-7.18	67.39	33.90	68.21	-7.19	27.57	55.83	-4.10
	7	0.18	0.28	1.50	-7.19	5.95	2.60	7.37	-7.19	96.76	48.54	97.94	-7.20	39.72	80.38	-4.11
	10	0.26	0.41	2.32	-7.19	9.43	3.97	11.52	-7.18	132.12	66.24	134.01	-7.20	54.47	110.50	-4.11
6,000	5	0.18	0.29	1.50	-7.53	6.76	2.84	8.10	-7.53	150.85	75.72	151.97	-7.54	43.59	86.94	-4.27
	7	0.25	0.41	2.26	-7.54	9.60	4.08	11.60	-7.53	217.78	109.29	219.51	-7.54	59.78	121.31	-4.28
	10	0.36	0.64	3.65	-7.54	15.08	6.34	18.53	-7.53	278.71	140.11	281.39	-7.54	77.67	157.14	-4.28
8,000	5	0.24	0.38	1.94	-7.77	9.37	3.92	11.01	-7.77	268.12	134.39	269.52	-7.77	56.34	113.49	-4.39
	7	0.33	0.54	3.04	-7.78	13.70	5.66	16.48	-7.77	370.03	185.47	372.28	-7.78	78.35	158.01	-4.40
	10	0.47	0.87	5.01	-7.77	20.83	8.67	25.29	-7.77	495.10	248.63	498.80	-7.77	103.44	209.54	-4.40
10,000	5	0.28	0.44	2.26	-7.96	13.27	5.38	15.29	-7.96	435.52	218.43	437.25	-7.97	72.18	145.72	-4.49
	7	0.40	0.66	3.72	-7.97	18.70	7.63	22.01	-7.97	616.33	309.09	619.10	-7.98	97.89	197.66	-4.50
	10	0.58	1.06	6.24	-7.97	27.54	11.26	32.94	-7.96	774.17	388.06	779.02	-7.97	130.14	264.09	-4.49

Notes: Parameters: $r = 0.06$, $\beta = 0.93$, $T = 20$, $\alpha = 0.77$, $\tau = 0.07$, $\kappa = 0.075$, $\iota = 0.01$, $A_{\max} = 40$, $H_{\max} = 5$, $w_{\max} = 40$, $P^r = 0.1$. Income: $\log y = z + \varepsilon$; $z_t = \rho z_{t-1} + \eta_t$ with $\rho = 0.977$, $\sigma_\eta = 0.024$, $\sigma_\varepsilon = 0.063$ on 7×7 grid. **U.Env**: average upper envelope time per period. **Cons.**: average homeowner consumption stage time per period. **Total**: average total solution time per period. **E.Err**: Euler equation error in mean $\log_{10}$ deviation. All time in seconds.

considerably larger than that in Section 2.1.4. At 500 grid points, FUES already reduces upper envelope time by roughly one order of magnitude relative to MSS (9 -10 times) and by 24-25 times relative to LTM. This advantage grows sharply with grid refinement: at 10,000 asset points, the upper envelope step takes only 282-577 ms under FUES, compared to 13-28 seconds (s) under MSS and 436-774 s under LTM - approximately 47-48 times faster than MSS and over 1,300-1,500 times faster than LTM (more than three orders of magnitude).

These upper envelope time savings translate into substantial reductions in the time required to solve the consumption stage. At 1,000 grid points, the homeowner consumption stage takes only 0.07-0.15 s per period under FUES, compared to 0.38-0.83 s under MSS, 2.0-4.1 s under LTM, and 6.7-13.1 s under VFI - yielding speedups of roughly five times over MSS, 27-29 times over LTM, and 90–95 times over VFI. At 10,000 points, FUES solves the consumption stage in just 0.44-1.06 s per period, compared to 5.4-11.3 s under MSS, 218-388 s under LTM, and 72-130 s under VFI - yielding speedups of roughly 11 times over MSS, 370-500 times over LTM, and 120-165 times over VFI. In multi-stage problems, the impact of improved EGM efficiency on total solution time thus depends on the relative computational burden of other stages within the problem. In our application, the consumption stage speed gains propagate through to deliver total time improvements of approximately 42-64 times relative to VFI, 5-7 times relative to MSS, and 125-193 times relative to LTM at the finest grids. The ranges in these speedup factors reflect variation in the housing grid dimension H: as H grid size rises, both MSS and LTM degrade more rapidly than FUES, amplifying the relative advantage of our method. Notably, while MSS remains faster than VFI for all grid sizes (though its relative advantage narrows at finer grids), LTM becomes *slower* than VFI once the W grid exceeds 4,000 points, and by 10,000 points it is roughly three times slower than VFI. In turn, increasing the W grid size from 1,000 to 10,000 points under FUES raises total computation time by only a factor between five and eight (depending on H grid), while still delivering substantial better accuracy.

These results show that in sufficiently rich models - where high-resolution grids yield substantially higher accuracy - the scaling properties of the upper envelope step can become the primary driver of end-to-end solution time. In such settings, EGM retains a decisive advantage over VFI only when paired with an efficient upper envelope routine. FUES effectively eliminates computational bottlenecks arising from discrete-continuous choices, making it practical to achieve large accuracy gains from finer grids at modest additional computational cost.

Model with capital income tax. We now introduce the piecewise-linear tax schedule (27), and compute the model in the presence of tax bracket boundaries and jump discontinuities as presented in Appendix Figure 9. As mentioned, the constraint segments at tax kinks preclude the use of MSS and LTM, leaving only FUES and VFI as viable solution methods. Table 5 shows that once we consider the capital income tax, FUES remains fast and accurate, while VFI becomes prohibitively expensive. Across all reported grid sizes, FUES solves the consumption stage in 0.21-1.24 s versus 11.31-109.14 s under VFI (i.e., roughly 50–90 times faster), reducing total per-period runtime from 22.93-220.38 s to 0.81-4.86 s (i.e., about 28–45 times). At the same time, it delivers Euler errors that are roughly 1.5–2.5 orders of magnitude smaller than VFI. These results confirm that in the presence of tax-induced kinks and notches - where standard monotonicity-based envelope routines are unusable - combining EGM with FUES restores solution tractability without sacrificing the EGM-level accuracy and speed.

Table 5: Application 3 - Discrete housing choice model with taxes: Speed & accuracy

Grid		FUES				VFI		
W	*H*	U.Env	Cons.	Total	E.Err	Cons.	Total	E.Err
1,000	3	0.15	0.21	0.81	-4.64	11.31	22.93	-3.01
	5	0.25	0.37	1.29	-4.62	18.72	37.86	-3.00
	7	0.34	0.52	1.72	-4.63	26.28	53.99	-2.99
2,000	3	0.18	0.30	1.17	-5.27	23.10	46.66	-3.23
	5	0.30	0.52	2.01	-5.26	38.22	78.39	-3.22
	7	0.42	0.72	2.66	-5.28	54.36	109.54	-3.22
4,000	3	0.27	0.47	1.92	-5.83	46.11	93.49	-3.44
	5	0.46	0.84	3.30	-5.82	76.97	155.45	-3.43
	7	0.65	1.24	4.86	-5.84	109.14	220.38	-3.43

Notes: Parameters: See Table 4. Tax schedule: See Figure 9. **U.Env**: average upper envelope time per period. **Cons.**: average homeowner consumption stage time per period. **Total**: average total solution time per period. **E.Err**: Euler equation error in mean $\log_{10}$ deviation. All time in seconds.

More broadly, by accommodating local non-convexities and non-monotone policy mappings, FUES avoids the need for problem-specific adjustments or global optimization routines that are computationally demanding. This allows researchers to work with realistic institutional settings without sacrificing computational speed or numerical precision. In this sense, FUES provides a practical approach for incorporating policy complexity into dynamic settings, expanding significantly the set of economic models that can be solved efficiently using envelope-based methods.

3 General formulation of FUES

We now formally define a general discrete-continuous optimization problem in reduced form, apply EGM and state FUES as a formal pseudo-code, and show how FUES recovers the optimal endogenous grid points. Intuitively, away from the secondary kinks, the upper envelope is locally attained by a single smooth, concave segment of the value correspondence (see Figure 2). When candidate points are ordered by the endogenous grid, concavity in (x, v)-space is equivalent to weakly decreasing secant slopes along a smooth region of the upper envelope. Along the smooth region, the optimal policy cannot jump. This is exactly the local geometry that FUES enforces via jump detection, correctly recovering the upper envelope of the linear approximations of value functions associated with the future discrete choices under mild regularity conditions and a sufficiently fine grid.

3.1 The general discrete-continuous optimization problem

Let the set $\mathbb{D}$ be a finite set, and let $\left\{G^{\mathbf{d}}\right\}_{\mathbf{d}\in\mathbb{D}}$ be a family of functions where each $G^{\mathbf{d}}$ is indexed by a fixed $\mathbf{d}$. Assume $G^{\mathbf{d}}$ satisfies $G^{\mathbf{d}}\colon X \times X' \to \mathbb{R} \cup \{-\infty\}$ for each $\mathbf{d}$, with $X \subset \mathbb{R}$ and $X' \subset \mathbb{R}$, and that X and X' are compact intervals. In a dynamic programming problem, X is the decision state space, X' is the continuation or post-state space, and each element in the choice set $\mathbb{D}$ will be a future stochastic sequence of history-dependent choices associated with the secondary kinks (recall Figure 1). $G^{\mathbf{d}}$ will thus be the maximand on the RHS of the Bellman equation, holding the future discrete choices $\mathbf{d}$ fixed.

Consider the problem of maximizing the upper envelope U of the choice-specific functions wrt. the continuous choice variable x'

$$V(x) = \max_{x'\in X'} \quad U(x, x'), \tag{39}$$

where

$$U(x, x') = \max_{\mathbf{d}\in\mathbb{D}} \quad G^{\mathbf{d}}(x, x'), \tag{40}$$

such that x is fixed. Next, define

$$\mathcal{I}(x, x') \;=\; \arg\max_{\mathbf{d}\in\mathbb{D}} \; G^{\mathbf{d}}(x, x')$$

as the optimal discrete choice holding the arguments of each $G^{\mathbf{d}}$ fixed. Additionally, define the optimal policy function as

$$\sigma(x) = \arg\max_{x' \in X'} \quad U(x, x'), \tag{41}$$

and the future choice-specific policy function and choice-specific value function as

$$\sigma^{\mathbf{d}}(x) = \arg\max_{x' \in X'} \quad G^{\mathbf{d}}(x, x'), \qquad Q^{\mathbf{d}}(x) = \max_{x' \in X'} \quad G^{\mathbf{d}}(x, x'). \tag{42}$$

The problem above should be viewed as describing the maximand of the Bellman equation *holding the current period discrete choice fixed*. In addition to the secondary kinks associated with switching $\mathbf{d}$ in $\mathbb{D}$, for a fixed $\mathbf{d}$, we also allow each $G^{\mathbf{d}}$ to feature kinks and notches by assuming that it is piecewise smooth. In particular, write int X' for the interior of the post-state space. Let $\underline{x}' := \inf X'$, $\bar{x}' := \sup X'$, and fix $\mathcal{B} = \{b_0 < \cdots < b_J\} \subset X'$ with $b_0 = \underline{x}'$ and $b_J = \bar{x}'$. Set $\mathcal{B}_- := \{b_0, \ldots, b_{J-1}\}$ and $\mathcal{B}_+ := \{b_1, \ldots, b_J\}$. For $b \in \mathcal{B}_-$ and $b \in \mathcal{B}_+$, define the one-sided derivatives

$$G^{\mathbf{d}}_{2,+}(x, b) := \lim_{\varepsilon \downarrow 0} G^{\mathbf{d}}_2(x, b + \varepsilon), \qquad G^{\mathbf{d}}_{2,-}(x, b) := \lim_{\varepsilon \downarrow 0} G^{\mathbf{d}}_2(x, b - \varepsilon),$$

where limits are taken within the adjacent interval(s).

Our first assumption concerns the existence and uniqueness of optima and requires $G^{\mathbf{d}}$ to be concave and differentiable over the brackets defined by $\mathcal{B}$. Recall that the proper domain of an extended real-valued function f defined on X, denoted by $\operatorname{dom} f$, is the set of all x such that $f(x) \in \mathbb{R}$. Since infeasibility is encoded by allowing $G^{\mathbf{d}}(x, x') = -\infty$, we state regularity on the proper domain. For an extended real-valued function $f \colon Y \to \mathbb{R} \cup \{-\infty\}$, define its proper domain as $\operatorname{dom} f := \{y \in Y : f(y) \in \mathbb{R}\}$. For $G^{\mathbf{d}} \colon X \times X' \to \mathbb{R} \cup \{-\infty\}$ and fixed $x \in X$, write $\operatorname{dom} G^{\mathbf{d}}(x, \cdot) := \{x' \in X' : (x, x') \in \operatorname{dom} G^{\mathbf{d}}\}$.

Assumption 1 (Existence and differentiability)**.** *We have:*

1. *For each $x \in X$ and $\mathbf{d} \in \mathbb{D}$, $Q^{\mathbf{d}}(x) < \infty$ and $\sigma^{\mathbf{d}}(x)$ is unique.*
2. *Moreover, for each $x \in \operatorname{int} X$, each $\mathbf{d} \in \mathbb{D}$ and each bracket interval (b_{j-1}, b_j), $G^{\mathbf{d}}$ is strictly concave and $x' \mapsto G^{\mathbf{d}}(x, x')$ is differentiable on $\operatorname{dom} G^{\mathbf{d}}(x, \cdot) \cap (b_{j-1}, b_j)$. In addition, the one-sided derivatives $G^{\mathbf{d}}_{2,+}(x, b)$ and $G^{\mathbf{d}}_{2,-}(x, b)$ exist in $\mathbb{R} \cup \{-\infty, +\infty\}$ for each $b \in \mathcal{B}_-$ and $b \in \mathcal{B}_+$, respectively, where limits are taken within $\operatorname{dom} G^{\mathbf{d}}(x, \cdot)$.*

Since we are holding $\mathbf{d}$ fixed, the second point in Assumption 1 is not restrictive for standard problems with occasionally binding constraints (Li and Stachurski, 2014; Dobrescu and Shanker, 2024). This implies that local optima satisfy the Euler equation. Formally:

Remark 1 (Conditional first-order inequalities)**.** *Fix $x \in X$ and $\mathbf{d} \in \mathbb{D}$. Under Assumption 1, the unique maximizer $\sigma^{\mathbf{d}}(x) \in X'$ of $x' \mapsto G^{\mathbf{d}}(x, x')$ satisfies*

$$\begin{aligned} \sigma^{\mathbf{d}}(x) \in \operatorname{int} X' \setminus (\mathcal{B}_- \cup \mathcal{B}_+) \;&\Rightarrow\; G_2^{\mathbf{d}}\Big(x, \sigma^{\mathbf{d}}(x)\Big) = 0, \\ \sigma^{\mathbf{d}}(x) \in \mathcal{B}_- \;&\Rightarrow\; G_{2,+}^{\mathbf{d}}\Big(x, \sigma^{\mathbf{d}}(x)\Big) \leq 0, \\ \sigma^{\mathbf{d}}(x) \in \mathcal{B}_+ \;&\Rightarrow\; G_{2,-}^{\mathbf{d}}\Big(x, \sigma^{\mathbf{d}}(x)\Big) \geq 0. \end{aligned} \tag{43}$$

We now detail the mapping between this general discrete-continuous optimization problem and the three applications in Section 2. Note that in all the applications below, we have $u(c) = -\infty$ if $c \leq 0$, which imposes the feasibility constraint in each case.

Application 1, Section 2.1. Fix t and consider agents deciding to continue working in $t+1$. Each $\mathbf{d}$, with $\mathbf{d} \in \mathbb{D}$, corresponds to a future path of feasible discrete choices made from $t+1$ to T. Formally, $\mathbb{D} = \{0,1\}^{T-t}$. The choice-specific payoff $G^{\mathbf{d}}(x, x')$ evaluates $u((1+r)x + y - x') + \beta V_{t+1}^{\mathbf{d}}(x')$, where x denotes financial assets, x' is the assets choice taken into the next period, and $V_{t+1}^{\mathbf{d}}$ is the continuation value function for those working in $t+1$ conditional on the future path of discrete choices made from $t+1$ to T. Since $x' \mapsto G^{\mathbf{d}}(x, x')$ is concave when $\mathbf{d}$ is held fixed, the boundary set $\mathcal{B}$ reduces to $\{\underline{x}', \bar{x}'\}$.

Application 2, Section 2.2. Fix t, recall our discussion in Section 2.2.3, and consider those deciding to adjust their housing at time t. Each $\mathbf{d}$, with $\mathbf{d} \in \mathbb{D}$, corresponds to a history-dependent stochastic path of housing adjustment choices from $t+1$ to T. The choice-specific payoff $G^{\mathbf{d}}$ corresponds to

$$G^{\mathbf{d}}(x, x') = u(x - \hat{\psi}_t^{\mathbf{d}}(x, x') - (1+\tau)x', x') + \beta \mathbb{E}_{y_t} V_{t+1}^{\mathbf{d}}(\hat{\psi}_t^{\mathbf{d}}(x, x'), x', y'),$$

where x is the total (financial and housing) wealth, x' represents the $t+1$ housing choice, $V_{t+1}^{\mathbf{d}}$ is the $t+1$ choice-specific value function, and $\mathbb{E}_{y_t}$ takes expectations over future income y' conditional on current income y_t. For a fixed discrete choice, the function $\hat{\psi}_t^{\mathbf{d}}$ maps the total wealth and housing choice to the optimal choice of financial assets taken into the next period, as defined by the Euler equation (17). Once again, the boundary set $\mathcal{B}$ reduces to $\{\underline{x}', \bar{x}'\}$.

Application 3, Section 2.3. Fix t and consider homeowners deciding to adjust housing at time t. Each $\mathbf{d}$, with $\mathbf{d} \in \mathbb{D}$, corresponds to a stochastic history-dependent path of housing choices and non-convex tax brackets from $t+1$ to T. The choice-specific payoff $G^{\mathbf{d}}$

corresponds to

$$G^{\mathbf{d}}(x, x') = u(x + H - x' - (1 + \tau)H', H') + \beta \mathbb{E}_{y_t} V^{\mathbf{d}}_{t+1}((1 + r)x' - T(x'), H', y'), \quad (44)$$

where x' represents financial assets taken into the next period, $V^{\mathbf{d}}_{t+1}$ is the next period choice-specific value function, $T(\cdot)$ is a (piecewise-linear) tax schedule on capital returns, and H and H' are the fixed beginning- and end-of-period housing assets. The boundary set $\mathcal{B}$ is the set of tax brackets and notches defined by (27).

3.1.1 EGM and formal FUES pseudo-code

Let $\hat{\mathbb{X}}'$ be the exogenous grid of continuation state values x', with $|\hat{\mathbb{X}}'| = N$, where N is the exogenous grid size. We assume grids are sequences, that is $\hat{\mathbb{X}}' = \{\hat{x}'_0, \hat{x}'_1 \dots \hat{x}'_i \dots\}$. We now define the *unrefined* grids produced by the EGM step. For each exogenous post-state point $\hat{x}'_i \in \hat{\mathbb{X}}'$, let $\hat{x}_i \in X$ satisfy the conditional FOCs in (43). Collect the resulting endogenous points in $\hat{\mathbb{X}} = \{\hat{x}_i\}_{i=0}^{N-1}$. Define the associated values by

$$\hat{v}_i := G^{\mathcal{I}(\hat{x}_i, \hat{x}'_i)}(\hat{x}_i, \hat{x}'_i), \qquad \hat{\mathbb{V}} := \{\hat{v}_i\}_{i=0}^{N-1}. \quad (45)$$

Moreover, let $\hat{z}_i \in \{1, \dots, J\}$ be the bracket index of the post-state $\hat{x}'_i$, defined by $\hat{z}_i = j$ if $\hat{x}'_i \in [b_{j-1}, b_j]$ for some $j \in \{1, \dots, J-1\}$. Collect these indices in a grid $\hat{\mathbb{Z}}$.

Three key observations regarding the definition of the endogenous grid values above should be noted. First, we require that $\mathcal{I}(\hat{x}_i, \hat{x}'_i)$ is optimal *given* $(\hat{x}_i, \hat{x}'_i)$, and that this does not imply $\mathcal{I}(\hat{x}_i, \hat{x}'_i)$ and $\hat{x}'_i$ solve (39). This holds for endogenous grids constructed for time consistent dynamic problems.[9] Second, we leave open the possibility that the endogenous grid $\hat{\mathbb{X}}$ is a general grid, *computed analytically or numerically*, of points that satisfy the necessary FOCs of a discrete-continuous optimization problem. Thus, $\hat{\mathbb{X}}'$ may contain multiple identical values. Third, feasible bracket-constrained solutions with $\hat{x}'_i \in \mathcal{B}$ that satisfy the corresponding one-sided inequality in (43) are included; in particular, sufficiently dense bracket-constrained candidates are part of the grid (Dobrescu and Shanker, 2024). As such, the bracket indicator $\hat{\mathbb{Z}}$ allows the scan to detect bracket switches even when the policy is locally flat (e.g. at tax-bracket boundaries as in Section 2.3.3).

The endogenous grid will be initially ordered in ascending order by the order of $\hat{\mathbb{X}}'$. We then order grids $\hat{\mathbb{X}}'$, $\hat{\mathbb{Z}}$, $\hat{\mathbb{V}}$, and $\hat{\mathbb{X}}$ by the values in $\hat{\mathbb{X}}$, and by a slight abuse of notation,

[9] We state our theoretical results for the time consistent case for tractability. However, FUES can also handle time inconsistent problems and extending the theory to time inconsistent problems is possible by defining G as a piecewise smooth function between all secondary kinks.

we continue to denote the reordered grids as $\hat{\mathbb{X}}'$, $\hat{\mathbb{Z}}$, $\hat{\mathbb{V}}$ and $\hat{\mathbb{X}}$. For each of these grids, FUES will generate a subsequence of refined grids $\mathbb{X}'$, $\mathbb{Z}$, $\mathbb{V}$ and $\mathbb{X}$ indexed by i_k - for instance, $\mathbb{X} = \left\{\hat{x}_0, \hat{x}_1, \ldots, \hat{x}_{i_k}, \ldots\right\}$. In the algorithm below, we let $N = |\hat{\mathbb{X}}|$ and $\bar{M}$ (i.e., the constant denoting the threshold at which FUES registers a jump between policy functions) be chosen by the researcher. Finally, $\mathcal{C}_{l+1}$ is a Boolean value that is true if point x_{l+1} is optimal, and $\mathcal{D}_l$ is a Boolean value that is true if x_k is not optimal and should be deleted.

Algorithm 1: Fast upper envelope scan

Data: $N > 0$, $\bar{M} > 0$, $\hat{\mathbb{X}}'$,$\hat{\mathbb{Z}}$, $\hat{\mathbb{V}}$,$\hat{\mathbb{X}}$
Result: $\mathbb{X}'$, $\mathbb{Z}$, $\mathbb{X}$ and $\mathbb{V}$

$k \leftarrow 1, l \leftarrow 1$;
$\hat{x}'_{i_q} \leftarrow \hat{x}'_q, \hat{z}_{i_q} \leftarrow \hat{z}_q, \hat{v}_{i_q} \leftarrow \hat{v}_q, \hat{x}_{i_q} \leftarrow \hat{x}_q \quad q \leq 1$; /* Init. first two points. */
while $l < N - 1$ **do**
  $g^v_k \leftarrow \frac{\hat{v}_{i_k} - \hat{v}_{i_{k-1}}}{\hat{x}_{i_k} - \hat{x}_{i_{k-1}}}, g^v_{l+1} \leftarrow \frac{\hat{v}_{l+1} - \hat{v}_{i_k}}{\hat{x}_{l+1} - \hat{x}_{i_k}}, g^x_{l+1} \leftarrow \frac{\hat{x}'_{l+1} - \hat{x}'_{i_k}}{\hat{x}_{l+1} - \hat{x}_{i_k}}$;
  $\mathcal{D}_l \leftarrow$ False ; /* Init. bwd. scan flag */
  if *If* $g^v_{l+1} \leq g^v_k$ *and* $\left(|g^x_{l+1}| > \bar{M} \vee \hat{z}_{l+1} \neq \hat{z}_{i_k}\right)$; /* Right turn, jump/ bracket switch */
  then
    $\mathcal{C}_{l+1} \leftarrow$ False;
    $\mathcal{C}_{l+1} \leftarrow$ Forward scan ; /* Optional fwd. scan (Fig. 5) */
  else if $g^v_{l+1} > g^v_k$; /* Left turn */
  then
    $\mathcal{C}_{l+1} \leftarrow$ True ;
    $\mathcal{D}_l \leftarrow$ Backward scan ; /* Optional bwd. scan (Fig. 5) */
  else
    $\mathcal{C}_{l+1} \leftarrow$ True ; /* Right turn, no jump */
  end
  if $\mathcal{D}_l$ **then**
    $\hat{x}'_{i_k} \leftarrow \hat{x}'_{l+1}, \hat{v}_{i_k} \leftarrow \hat{v}_{l+1}$;
    $\hat{x}_{i_k} \leftarrow \hat{x}_{l+1}, \hat{z}_{i_k} \leftarrow \hat{z}_{l+1}$; /* Replace i_k subopt. point. */
  else if $\mathcal{C}_{l+1}$ **then**
    $\hat{x}'_{i_{k+1}} \leftarrow \hat{x}'_{l+1}, \hat{v}_{i_{k+1}} \leftarrow \hat{v}_{l+1}$;
    $\hat{x}_{i_{k+1}} \leftarrow \hat{x}_{l+1}, \hat{z}_{i_{k+1}} \leftarrow \hat{z}_{l+1}$; /* Add point; keep i_k opt. point. */
    $k = k + 1$
  else
    Pass
  end
  $l \leftarrow l + 1$;
end

3.2 The main accuracy result

Assumptions. Define the optimal subset of points $\mathbb{X}^\star$ of the endogenous grid as

$$\mathbb{X}^\star : = \left\{ \hat{x}_i \in \hat{\mathbb{X}} \,\middle|\, \hat{v}_i = V(\hat{x}_i) \right\}, \tag{46}$$

and assume that the distance between choice-specific policy functions is bounded below. This assumption is satisfied under the condition that future-choice specific policy functions do not intersect and that there are finitely many future choices in $\mathbb{D}$. In applications, this holds when (i) there are finitely many periods, (ii) the shock grid has finite states, and (iii) for each decision state, the current discrete choice is unique. Formally:

Assumption 2 (Minimum jump size)**.** *There exists a constant $D > 0$ such that for all distinct $\mathbf{d}, \mathbf{s} \in \mathbb{D}$ and all $x \in X$ with $\sigma^{\mathbf{d}}(x), \sigma^{\mathbf{s}}(x) \in \operatorname{int} X' \setminus \mathcal{B}$, we have $\left|\sigma^{\mathbf{d}}(x) - \sigma^{\mathbf{s}}(x)\right| \geq D$.*

Third, we assume that, within smooth regions of the maximand, the rate of change of the continuation states mapped by the transition and policy functions is uniformly bounded above by a common constant. This assumption is straightforward to satisfy in standard economic applications, where the policy function, conditional on a future stochastic sequence of discrete choices ($\sigma^{\mathbf{d}}$) constrained over a bracket, has a Lipschitz bound. This bound is commonly met since $\sigma^{\mathbf{d}}$ is the solution to a concave optimization problem, in which the optimal policy function exhibits well-defined maximal and minimal marginal propensities to consume (White, 2015; Carroll and Shanker, 2025). Crucially, this is a regularity conditional on the current bracket and future discrete choice. Thus, it does not restrict the global policy from featuring jumps or non-monotonicities when crossing bracket boundaries or switching $\mathbf{d}$.

Assumption 3 (Bracket-wise bi-Lipschitz monotonicity)**.** *For each $\mathbf{d} \in \mathbb{D}$ and each bracket interval $[b_{j-1}, b_j]$, define*

$$\mathcal{X}_j^{\mathbf{d}} := \left\{ x \in X : \sigma^{\mathbf{d}}(x) \in [b_{j-1}, b_j] \right\}, \qquad \mathcal{X}_j^{\mathbf{d},\circ} := \left\{ x \in X : \sigma^{\mathbf{d}}(x) \in (b_{j-1}, b_j) \right\}.$$

There exist common constants $M, M_{\text{inv}} < \infty$ such that, for every $(\mathbf{d}, j)$, $\sigma^{\mathbf{d}}$ is monotone (non-decreasing or non-increasing) and M-Lipschitz on $\mathcal{X}_j^{\mathbf{d}}$, and $(\sigma^{\mathbf{d}}|_{\mathcal{X}_j^{\mathbf{d},\circ}})^{-1}$ exists and is M_{inv}-Lipschitz on $\sigma^{\mathbf{d}}(\mathcal{X}_j^{\mathbf{d},\circ}) \subset (b_{j-1}, b_j)$.

Finally, we assume each pair $Q^{\mathbf{d}}$ and $Q^{\mathbf{s}}$ either never intersect or cross exactly once in X. A sufficient condition for this assumption to be satisfied is prudence - i.e., $u'''(x) > 0$, where u is per-period utility. While more general conditions are also possible, stan-

dard applications in heterogeneous agent macroeconomics and structural microeconometrics easily meet this requirement (Kimball, 1990; Carroll and Kimball, 1996; Carroll, 2006; Carroll et al., 2021). For any distinct pair $(\mathbf{d}, \mathbf{s}) \in \mathbb{D} \times \mathbb{D}$ with $\mathbf{d} \neq \mathbf{s}$, define $\Delta_{\mathbf{d},\mathbf{s}}(x) := Q^{\mathbf{d}}(x) - Q^{\mathbf{s}}(x)$ and then formally:

Assumption 4 (Single crossing of value functions)**.** *For every distinct pair $(\mathbf{d}, \mathbf{s}) \in \mathbb{D} \times \mathbb{D}$, either $Q^{\mathbf{d}}$ and $Q^{\mathbf{s}}$ never intersect on X, or they cross exactly once on X. Equivalently, one of the following holds:*

$$\begin{aligned}
&\textit{either} && \Delta_{\mathbf{d},\mathbf{s}}(x) > 0 && \textit{for all } x \in X, \\
&\textit{or} && \Delta_{\mathbf{d},\mathbf{s}}(x) < 0 && \textit{for all } x \in X, \\
&\textit{or} && \exists\, \bar{x} \in X :\ \Delta_{\mathbf{d},\mathbf{s}}(\bar{x}) = 0, && \\
& && \Delta_{\mathbf{d},\mathbf{s}}(x) > (<) 0 && \textit{for all } x \in X \cap (-\infty, \bar{x}), \\
& && \Delta_{\mathbf{d},\mathbf{s}}(x) < (>) 0 && \textit{for all } x \in X \cap (\bar{x}, \infty).
\end{aligned} \tag{47}$$

Theoretical results. Let N be the number of uniform points in the exogenous grid and let the above Assumptions 1-4 hold for the following results. Our first intermediate result tells us that when the exogenous grid is sufficiently fine, the distance between any two consecutive optimal points on the exogenous grid is bounded above by a constant.

Claim 1. *There exist $\epsilon > 0$ and N large enough such that for all $x^\star_{j+1}, x^\star_j \in \mathbb{X}^\star$, we have $|x^\star_{j+1} - x^\star_j| \leq \epsilon$ and $\frac{D}{\epsilon} > 2M$.*

Proof. See Appendix C.

An implication of the above claim is that for a sufficiently fine grid, the jump detection threshold can accurately detect jumps between future choice-specific policy functions.

Corollary 1. *For N large enough, there exists a jump detection threshold that satisfies $\frac{D}{\epsilon} - M \geq \bar{M} > M$.*

Proof. By Claim 1, choose $\epsilon > 0$ such that $\frac{D}{\epsilon} > 2M$. Set $\bar{M} := \frac{D}{2\epsilon}$. Then $\bar{M} > M$ and $\frac{D}{\epsilon} - M \geq \bar{M}$. □

To make explicit that the upper envelope also ranges over current bracket-constrained regions (e.g. tax brackets), define bracket-restricted conditional policy and value functions. For each $\mathbf{d} \in \mathbb{D}$ and each interval index $j \in \{1, \ldots, J\}$, define

$$Q^{\mathbf{d}}_j(x) := \max_{x' \in [b_{j-1}, b_j]} G^{\mathbf{d}}(x, x'), \qquad x \in X,$$

with the convention $Q_j^{\mathbf{d}}(x) = -\infty$ if $G^{\mathbf{d}}(x, x') = -\infty$ for all $x' \in [b_{j-1}, b_j]$. Let

$$X_j^{\mathbf{d}} := \{x \in X : Q_j^{\mathbf{d}}(x) > -\infty\}, \qquad \sigma_j^{\mathbf{d}}(x) := \arg \max_{x' \in [b_{j-1}, b_j]} G^{\mathbf{d}}(x, x') \quad (x \in X_j^{\mathbf{d}}),$$

and extend $\sigma_j^{\mathbf{d}}$ arbitrarily to $X \setminus X_j^{\mathbf{d}}$. Each non-empty $\sigma_j^{\mathbf{d}}$ defines a continuous segment of the conditional policy mapping associated with $\mathbf{d}$ constrained by a bracket, and it may also include boundary (constrained) points. Let

$$\bar{\mathbb{D}} := \{(\mathbf{d}, j) : \mathbf{d} \in \mathbb{D},\ j \in \{1, \ldots, J\},\ X_j^{\mathbf{d}} \neq \emptyset\}$$

be the resulting set of segments. To ease notation in what follows, we relabel elements of $\bar{\mathbb{D}}$ by $\mathbf{d}$ and continue to write $\mathbb{D}$ and $Q^{\mathbf{d}}$ for this expanded set of segments. Next, for each $\mathbf{d} \in \mathbb{D}$, define $\mathbb{X}^{\mathbf{d}} \subset \hat{\mathbb{X}}$ by

$$\mathbb{X}^{\mathbf{d}} \ := \ \{\hat{x}_i \in \hat{\mathbb{X}} : \hat{v}_i = Q^{\mathbf{d}}(\hat{x}_i)\},$$

and let $\widetilde{Q}^{\mathbf{d}}$ be the unique piecewise-linear function on the convex hull of $\mathbb{X}^{\mathbf{d}}$ that interpolates $Q^{\mathbf{d}}$ on $\mathbb{X}^{\mathbf{d}}$. Set the upper envelope of the linear interpolants as

$$\widetilde{U}(x) \ = \ \max_{\mathbf{d} \in \mathbb{D}} \widetilde{Q}^{\mathbf{d}}(x),$$

and define the set of points on the upper envelope of interpolants by

$$\widetilde{\mathbb{X}}^{\star} \ := \ \{\hat{x}_i \in \hat{\mathbb{X}} : \hat{v}_i = \widetilde{U}(\hat{x}_i)\}. \tag{48}$$

For our main result below, let $(\mathbb{X}, \mathbb{X}', \mathbb{Z}, \mathbb{V})$ be the tuple of outputs of Algorithm 1 with the forward and backward scans.

Proposition 1. *Let Assumptions 1-4 hold. There exists a grid size N large enough such that there exists a jump detection threshold $\bar{M}$ with $\frac{D}{\epsilon} - M \geq \bar{M} > M$ and $\hat{v}_i = \widetilde{U}(\hat{x}_i)$ for each $\hat{x}_i \in \mathbb{X}$.*

For an exogenous grid dense enough, the values $\hat{v}_i$ recovered by FUES will thus be on the upper envelope of the linear interpolants of segments generated by the future discrete choices (that cause the secondary kinks) and contemporaneous non-smooth brackets.

3.3 The proof of the main result

Let $T_P \subset X$ be the set of *crossing points* between the segment-specific value functions - i.e., the set of $x \in X$ such that for some $\mathbf{m}, \mathbf{q} \in \mathbb{D}$, $Q^{\mathbf{m}}(x) = Q^{\mathbf{q}}(x)$ and for some $\epsilon > 0$,

$V(x+\epsilon) = Q^{\mathbf{q}}(x+\epsilon) > Q^{\mathbf{m}}(x+\epsilon)$ and $V(x-\epsilon) = Q^{\mathbf{m}}(x-\epsilon) > Q^{\mathbf{q}}(x-\epsilon)$. For each $\mathbf{d} \in \mathbb{D}$, define the $\mathbf{d}$-optimal region and its restriction to the endogenous grid by $X^{\mathbf{d}} := \{x \in X : V(x) = Q^{\mathbf{d}}(x)\}$ and $\widetilde{X}^{\mathbf{d}} := X^{\mathbf{d}} \cap \hat{\mathbb{X}}$.

Proof of Proposition 1. We prove Proposition 1 by showing

$$\hat{x}_i \in \mathbb{X} \iff \hat{x}_i \in \widetilde{\mathbb{X}}^{\star}, \tag{49}$$

for all $i \leq |\hat{\mathbb{X}}|$. In particular, we will show that if (49) holds for all i, with $i \leq l$ for some $l \leq |\hat{\mathbb{X}}|$, then it will hold for all i with $i \leq l+1$. Thus, by the principle of induction, the claim will hold for all i with $i \leq l$ for any l including $l = |\hat{\mathbb{X}}|$.

Induction start. For the inductive argument below, it suffices to start from any pair of retained points $(\hat{x}_{i_0}, \hat{x}_{i_1})$ with $\hat{x}_{i_0} < \hat{x}_{i_1}$ and $\hat{x}_{i_0}, \hat{x}_{i_1} \in \widetilde{\mathbb{X}}^{\star}$. We verify at the end of the proof that Algorithm 1, although seeded with $(\hat{x}_0, \hat{x}_1)$, reaches such a starting pair via the backward-scan replacement step.

Fix l, with $l \leq |\hat{\mathbb{X}}|$, and make the inductive assumption that (49) holds for all i, with $i \leq l$. Note that $\mathbb{X}$ is a subsequence of $\hat{\mathbb{X}}$, which we can denote as $\mathbb{X} = \{\hat{x}_{i_j}\}_{j=0}^{N_F}$, where N_F is the number of points recovered by the FUES algorithm. As such, for any k such that $i_k \leq l$, let $\{\hat{x}_{i_0}, \ldots, \hat{x}_{i_j}, \ldots, \hat{x}_{i_k}\}$ denote the first k points of $\mathbb{X}$.

Part 1: Proof that $\hat{x}_i \in \mathbb{X} \implies \hat{x}_i \in \widetilde{\mathbb{X}}^{\star}$ holds for all i, with $i \leq l+1$.

For this part of the proof, fix k such that $k = \max\{j : i_j \leq l\}$. If $i_{k+1} > l+1$, then the proof of this part is complete since $\hat{x}_i \in \mathbb{X} \implies \hat{x}_i \in \widetilde{\mathbb{X}}^{\star}$ holds for all $i < l+1$ by the inductive assumption. On the other hand, suppose $i_{k+1} = l+1$. We wish to now show that $\hat{x}_{i_{k+1}} \in \widetilde{\mathbb{X}}^{\star}$ (where the point $\hat{x}_{i_{k+1}}$ satisfies $\hat{x}_{i_{k+1}} \in \mathbb{X}$). There are two cases. The first case arises if a right turn is made on the value correspondence from the point $\hat{x}_{i_k}$, $\frac{\hat{v}_{i_{k+1}} - \hat{v}_{i_k}}{\hat{x}_{i_{k+1}} - \hat{x}_{i_k}} \leq \frac{\hat{v}_{i_k} - \hat{v}_{i_{k-1}}}{\hat{x}_{i_k} - \hat{x}_{i_{k-1}}}$. The second case arises if a left turn is made on the value correspondence from the point $\hat{x}_{i_k}$, $\frac{\hat{v}_{i_{k+1}} - \hat{v}_{i_k}}{\hat{x}_{i_{k+1}} - \hat{x}_{i_k}} > \frac{\hat{v}_{i_k} - \hat{v}_{i_{k-1}}}{\hat{x}_{i_k} - \hat{x}_{i_{k-1}}}$.

[Part 1, Case 1: Right turn]

Suppose $\hat{v}_{i_k} = Q^{\mathbf{m}}(\hat{x}_{i_k})$ for a future sequence of discrete choices $\mathbf{m}$, and let $\hat{v}_{i_{k+1}} = Q^{\hat{\mathbf{m}}}(\hat{x}_{i_{k+1}})$, where $\hat{\mathbf{m}}$ may or may not be the same as $\mathbf{m}$. We divide the analysis into two sub-cases.

Sub-case (i): The point $\hat{x}_{i_{k+1}}$ leads to the same future choice sequence as $\hat{x}_{i_k}$ and $\hat{\mathbf{m}} = \mathbf{m}$.

If $\hat{x}_{i_{k+1}} \in \mathbb{X}^\star$, then $\hat{x}_{i_{k+1}}$ must also belong to $\widetilde{\mathbb{X}}^\star$ (Claim 2 in Appendix C). On the other hand, if $\hat{x}_{i_{k+1}} \notin \mathbb{X}^\star$, it must be that $\hat{x}_{i_{k+1}} \notin X^{\mathbf{m}}$. Thus, $\hat{x}_{i_{k+1}}$ is strictly greater than the smallest crossing point greater than $\hat{x}_{i_k}$, which we denote by x^{TP}. Now, let $\hat{x}^\dagger > x^{\text{TP}}$ be the first optimal point in $\mathbb{X}^\star$ strictly greater than x^{TP}. The point $\hat{x}^\dagger$ leads the optimal future choice sequence $\mathbf{q} \neq \mathbf{m}$. If $\hat{x}^\dagger$ is the minimal element in $\widetilde{X}^{\mathbf{q}} \cap \mathbb{X}$, let $\hat{x}^{\dagger+1} \in \widetilde{X}^{\mathbf{q}} \cap \mathbb{X}$ be the next grid point on the same future discrete choice sequence $\mathbf{q}$. Now, extrapolate the line segment from $\left(\hat{x}^\dagger,\, Q^{\mathbf{q}}(\hat{x}^\dagger)\right)$ to $\left(\hat{x}^{\dagger+1},\, Q^{\mathbf{q}}(\hat{x}^{\dagger+1})\right)$ to the point $\hat{x}_{i_{k+1}} < \hat{x}^\dagger$ to form the evaluation of $\widetilde{Q}^{\mathbf{q}}(\hat{x}_{i_{k+1}})$. However, the backward scan procedure eliminates $\hat{x}_{i_{k+1}}$ from the subsequence $\mathbb{X}$ if $\widetilde{Q}^{\mathbf{q}}(\hat{x}_{i_{k+1}}) > \widetilde{Q}^{\mathbf{m}}(\hat{x}_{i_{k+1}}) = \hat{v}_{i_{k+1}}$. This implies that $\widetilde{Q}^{\mathbf{q}}(\hat{x}_{i_{k+1}}) \leq \widetilde{Q}^{\mathbf{m}}(\hat{x}_{i_{k+1}}) = \hat{v}_{i_{k+1}}$ and thus $\hat{x}_{i_{k+1}} \in \widetilde{\mathbb{X}}^\star$.

If $\hat{x}^\dagger$ is not the minimal element in $\widetilde{X}^{\mathbf{q}} \cap \mathbb{X}$, then let $\hat{x}^{\dagger-1} \in \widetilde{X}^{\mathbf{q}} \cap \mathbb{X}$ be the grid point on the same future discrete choice sequence $\mathbf{q}$ to the left of $\hat{x}^\dagger$. Note that this point must be to the left of x^{TP}. Thus, the line segment from $\left(\hat{x}^\dagger,\, Q^{\mathbf{q}}(\hat{x}^\dagger)\right)$ to $\left(\hat{x}^{\dagger-1},\, Q^{\mathbf{q}}(\hat{x}^{\dagger-1})\right)$ can be evaluated at the point $\hat{x}_{i_{k+1}} < \hat{x}^\dagger$ to evaluate $\widetilde{Q}^{\mathbf{q}}(\hat{x}_{i_{k+1}})$. Hence, by the backward scan, if $\widetilde{Q}^{\mathbf{q}}(\hat{x}_{i_{k+1}})$ evaluates strictly above $\hat{v}_{i_{k+1}}$, then $\hat{x}_{i_{k+1}}$ is removed as suboptimal. Otherwise, if $\widetilde{Q}^{\mathbf{q}}(\hat{x}_{i_{k+1}})$ is below $\hat{v}_{i_{k+1}}$, the point remains in $\mathbb{X}$ (i.e., $\hat{x}_{i_{k+1}} \in \mathbb{X}$) and must lie on the envelope, so $\hat{x}_{i_{k+1}} \in \widetilde{\mathbb{X}}^\star$.

Sub-case (ii): The point $\hat{x}_{i_{k+1}}$ switches to a distinct future sequence $\hat{\mathbf{m}} \neq \mathbf{m}$.

If $\hat{\mathbf{m}} \neq \mathbf{m}$ and $\hat{x}_{i_{k+1}} \leq x^{\text{TP}}$, then by Claim 5, the FUES algorithm detects a discontinuous jump (or a bracket switch of z values) and removes $\hat{x}_{i_{k+1}}$. The possibility left to consider is that $\hat{x}_{i_{k+1}} > x^{\text{TP}}$. In this case, the FUES algorithm also detects a discontinuous jump, and if $\hat{x}_{i_{k+1}}$ remains in $\mathbb{X}$, the forward scan must not have eliminated it. Specifically, the forward scan searches to the right of $\hat{x}_{i_{k+1}}$ for a point $\hat{x}^\dagger$ on the same future discrete choice sequence as $\hat{x}_{i_k}$ (i.e., $\mathbf{m}$), and checks whether $(\hat{x}_{i_{k+1}}, \hat{v}_{i_{k+1}})$ dominates the segment joining $(\hat{x}_{i_k}, \hat{v}_{i_k})$ and $(\hat{x}^\dagger, Q^{\mathbf{m}}(\hat{x}^\dagger))$ (cf. the forward scan description above). If such a $\hat{x}^\dagger$ exists, the algorithm compares the slope from $(\hat{x}_{i_k}, \hat{v}_{i_k})$ to $(\hat{x}^\dagger, Q^{\mathbf{m}}(\hat{x}^\dagger))$ with the slope from $(\hat{x}_{i_k}, \hat{v}_{i_k})$ to $(\hat{x}_{i_{k+1}}, \hat{v}_{i_{k+1}})$. If the former slope is strictly greater, then $\hat{x}_{i_{k+1}}$ lies below the segment and is deemed suboptimal and removed. If the former slope is less than the latter, then $\hat{x}_{i_{k+1}}$ dominates that segment and is retained (i.e., $\hat{x}_{i_{k+1}} \in \mathbb{X}$). If no such $\hat{x}^\dagger$ is found, the forward scan cannot certify dominance and $\hat{x}_{i_{k+1}}$ is not an element of $\mathbb{X}$.

[Part 1, Case 2: Left turn]

Let $\frac{\hat{v}_{i_{k+1}} - \hat{v}_{i_k}}{\hat{x}_{i_{k+1}} - \hat{x}_{i_k}} > \frac{\hat{v}_{i_k} - \hat{v}_{i_{k-1}}}{\hat{x}_{i_k} - \hat{x}_{i_{k-1}}}$. Then by Claim 4 in the Appendix, we have $\hat{x}_{i_{k+1}} \in \widetilde{\mathbb{X}}^\star$.

Part 2: Proof that $\hat{x}_i \in \mathbb{X} \impliedby \hat{x}_i \in \widetilde{\mathbb{X}}^\star$ holds for i with $i \leq l+1$.

Now we show that if a candidate point $\hat{x}_{l+1}$ satisfies $\hat{x}_{l+1} \in \widetilde{\mathbb{X}}^{\star}$, then $\hat{x}_{l+1} \in \mathbb{X}$. Consider the first case of a right turn.

[Part 2, Case 1: Right turn]

Suppose $\hat{x}_{l+1}$ generates a right turn wrt. the line segment joining $\hat{x}_{i_{k-1}}$ and $\hat{x}_{i_k}$. Formally, let

$$\frac{\hat{v}_{l+1} - \hat{v}_{i_k}}{\hat{x}_{l+1} - \hat{x}_{i_k}} \;\leq\; \frac{\hat{v}_{i_k} - \hat{v}_{i_{k-1}}}{\hat{x}_{i_k} - \hat{x}_{i_{k-1}}}, \tag{50}$$

where $\hat{v}_{i_k} = Q^{\mathbf{m}}(\hat{x}_{i_k})$ for some future choice sequence $\mathbf{m} \in \mathbb{D}$. In addition, let $\hat{v}_{l+1} = Q^{\mathbf{q}}(\hat{x}_{l+1})$ for $\mathbf{q} \in \mathbb{D}$. The aim is to demonstrate that $\hat{x}_{l+1} \in \mathbb{X}$. There are two sub-cases to examine, $\mathbf{q} = \mathbf{m}$ and $\mathbf{q} \neq \mathbf{m}$.

Sub-case (i): $\mathbf{q} = \mathbf{m}$.

Fix N large enough that consecutive endogenous grid points lie at most ϵ apart, so in particular $\left|\hat{x}_{l+1} - \hat{x}_{i_k}\right| \leq \epsilon$. We now show that no policy jump can occur if both points pertain to the same discrete choice $\mathbf{m}$. To see why, denote

$$\hat{x}'_{l+1} \;=\; \sigma^{\mathbf{m}}(\hat{x}_{l+1}), \quad \hat{x}'_{i_k} \;=\; \sigma^{\mathbf{m}}(\hat{x}_{i_k}).$$

Under Assumption 3, it must hold that

$$\left|\hat{x}'_{l+1} - \hat{x}'_{i_k}\right| \;\leq\; M \left|\hat{x}_{l+1} - \hat{x}_{i_k}\right|, \tag{51}$$

which then implies

$$\frac{\left|\hat{x}'_{l+1} - \hat{x}'_{i_k}\right|}{\left|\hat{x}_{l+1} - \hat{x}_{i_k}\right|} \;\leq\; M < \bar{M}, \tag{52}$$

where the second inequality follows from Corollary 1. Thus, FUES does not remove the point $\hat{x}_{l+1}$ from $\mathbb{X}$, and $\hat{x}_{l+1} \in \mathbb{X}$ as was to be shown.

Sub-case (ii): $\mathbf{q} \neq \mathbf{m}$.

Here, $\hat{x}_{i_{k-1}}$ and $\hat{x}_{i_k}$ both lie in the region of the state space that yields the optimal future choice sequence $\mathbf{m}$, but $\hat{x}_{l+1}$ satisfies $V(\hat{x}_{l+1}) = Q^{\mathbf{q}}(\hat{x}_{l+1})$ for some $\mathbf{q} \neq \mathbf{m}$. Under the forward scan criterion, $\hat{x}_{l+1}$ can be retained only if $(\hat{x}_{l+1}, \hat{v}_{l+1})$ lies on the segment $\mathbf{m}$ and (weakly) above the segment joining $(\hat{x}_{i_k}, \hat{v}_{i_k})$ and a forward point $(\hat{x}^{\dagger}, Q^{\mathbf{m}}(\hat{x}^{\dagger}))$. Since $\hat{x}_{l+1} \in \widetilde{\mathbb{X}}^{\star}$, this dominance condition is satisfied, and the forward scan thus retains $\hat{x}_{l+1}$ (i.e., $\hat{x}_{l+1} \in \mathbb{X}$).

[Part 2, Case 2: Left turn]

Let us now assume that

$$\frac{\hat{v}_{l+1} - \hat{v}_{i_k}}{\hat{x}_{l+1} - \hat{x}_{i_k}} > \frac{\hat{v}_{i_k} - \hat{v}_{i_{k-1}}}{\hat{x}_{i_k} - \hat{x}_{i_{k-1}}}. \tag{53}$$

Under this condition, a standard slope comparison indicates that point $\hat{x}_{l+1}$ produces a strictly *higher* secant slope than the line segment from $(\hat{x}_{i_{k-1}}, \hat{v}_{i_{k-1}})$ to $(\hat{x}_{i_k}, \hat{v}_{i_k})$. By line 10 of Algorithm 1, the procedure interprets this higher slope as an upward or 'left turn' increase in the value function. Consequently, $\hat{x}_{l+1}$ is retained and $\hat{x}_{l+1} \in \mathbb{X}$.

Initialization. It remains to justify that the induction may be started from two points in $\widetilde{\mathbb{X}}^\star$ even though Algorithm 1 is seeded with $(\hat{x}_0, \hat{x}_1)$. Let $\hat{x}_j$ be the smallest grid point to the right of $\hat{x}_0$ such that $\hat{x}_j \in \widetilde{\mathbb{X}}^\star$ (for N large, such a j exists and $\widetilde{\mathbb{X}}^\star$ contains at least two points). If $j = 1$ there is nothing to show. Otherwise, by definition of j, every point $\hat{x}_i$ with $0 < i < j$ satisfies $\hat{v}_i < \widetilde{U}(\hat{x}_i)$, and since $\widetilde{U}$ is linear on $(\hat{x}_0, \hat{x}_j)$ (no envelope node lies between them), each such point lies strictly below the segment joining $(\hat{x}_0, \hat{v}_0)$ and $(\hat{x}_j, \hat{v}_j)$. Consequently, when the scan reaches $l + 1 = j$, the left turn segment is triggered and the backward scan dominance check (i.e., the same segment dominance logic used above in the backward scan cases) flags the currently stored intermediate point as suboptimal and replaces it by $\hat{x}_j$ via the update step *Replace i_k subopt. point* in Algorithm 1. Thus, after processing index j, the algorithm's active pair of retained points is $(\hat{x}_0, \hat{x}_j) \subset \widetilde{\mathbb{X}}^\star$, providing the required starting pair for the induction.

This completes the inductive step, and hence (49) holds for all $i \leq |\hat{\mathbb{X}}|$. Therefore, $\hat{v}_i = \widetilde{U}(\hat{x}_i)$ for each point retained by the algorithm, proving Proposition 1. □

4 Conclusions

We introduce a *fast upper envelope scan* (FUES) method for solving discrete-continuous stochastic dynamic models that admit a one-dimensional state representation. FUES efficiently constructs the upper envelope of the value correspondence generated by the EGM, overcoming a central limitation of existing envelope-based methods in settings with non-monotone policy functions. By eliminating the monotonicity requirement, FUES enables envelope-based solution methods in classes of models that previously failed, required substantial simplification or had to rely on slow global optimization routines.

We demonstrate the applicability and performance of FUES using three benchmark models drawn from the literature. Across these applications, FUES consistently outperforms

existing upper-envelope methods - including monotone segment selection (MSS), local triangulation methods (LTM), and the rooftop cut (RFC) - as well as hybrid approaches that rely on numerical optimization, such as nested EGM (NEGM). In settings where envelope-based methods are applicable, FUES typically delivers speed improvements on the order of magnitude while maintaining comparable solution accuracy. In environments where non-monotonicity prevents the use of existing envelope methods, FUES continues to recover the correct envelope and achieves over two orders of magnitude lower Euler equation errors than optimization-based alternatives. Importantly, these gains persist as model complexity increases and endogenous grids become more irregular due to uncertainty or non-monotonicity of policies.

The paper also establishes formal conditions under which FUES recovers the true upper envelope of the EGM value correspondence. In particular, we show that, under standard regularity assumptions - concavity and smoothness of value functions between kink points, bounded separation of policy jumps, and a sufficiently fine exogenous grid - FUES recovers the upper envelope of the piecewise-linear interpolants of the choice-specific value functions associated with different future-choice segments. Away from jump (crossing) regions - where the upper envelope coincides locally with a single concave segment - FUES introduces no additional approximation error beyond that inherent in the underlying EGM and interpolation steps: it selects the correct segment and recovers the corresponding envelope points exactly on that smooth region. The key requirement is that the exogenous grid is fine enough to 'catch' the secondary jumps in the continuation state policy function. Near jump points, however, the envelope is non-differentiable and the location of the kink is only identified up to grid resolution. No upper-envelope routine can sharpen this localization without introducing additional candidate points through grid refinement. Finally, the finite horizon and finite shock space restrictions ensure that the set of relevant future choice paths is finite and, consequently, that only finitely many policy jumps can occur and be detected.

While FUES can be applied as-is in infinite horizon settings, the possibility of infinitely many (and arbitrarily small) jumps cannot be ruled out generally; extending the theoretical guarantees to environments with infinitely many (and arbitrarily small) potential policy jumps or continuous shock distributions remains a promising direction for future work. In addition, FUES is designed for problems in which candidate EGM points can be ordered along a one-dimensional state space, a requirement that underpins the scan-based construction of the envelope. This approach is not yet feasible in higher dimensions, where endogenous grids are irregular and lack a natural order for scanning. An-

other promising area for future research is thus extending FUES to multiple dimensions by incorporating techniques such as Hilbert curves or other space-filling methods to produce an ordering of irregular multidimensional candidate points.

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Online Appendix

A Existing methods and their limitations

This section provides a brief overview of the main upper envelope algorithms in the literature, outlines their limitations, and explains how FUES overcomes them. Below we focus on three key drawbacks of alternative methods: (i) the inability to handle continuation state policies that are not strictly monotone, (ii) computational complexity, and (iii) the lack of any theoretical guarantees of solution accuracy. For completeness, we also provide the detailed pseudo-code for MSS (Iskhakov et al., 2017) and LTM (Druedahl and Jørgensen, 2017) since our focus remains on upper envelope methods. To the best of our knowledge, Fella (2014) was the first to detail the multiple-optima problem in discrete-continuous EGM and propose a solution. This solution relies, however, on local numerical optimization in regions where choice-specific value functions overlap, rather than on a dedicated upper envelope routine.

Monotone segment selection (MSS). A prominent approach to constructing upper envelopes generated by DC-EGM problems is MSS by Iskhakov et al. (2017). MSS exploits the fact that if the continuation state policy function (i.e., the function defining the optimal value of a_{t+1} in Application 1 above) is strictly monotone, then monotone segments that overlap to generate multiple optima can be separately identified by detecting where the endogenous grid decreases along the exogenous grid. Within each such segment, a single future sequence of discrete choices is implied. The algorithm interpolates the value function on every segment separately and then evaluates all interpolants on a common grid to assemble the upper envelope.

The first drawback of MSS is that complexity may increase as the number of overlapping future choice-specific value functions grows. This is because as the number of jumps grows, there will be a larger number of distinct segments that must be identified, stored as separate objects, and then interpolated over. As we show below, MSS computation times also scale linearly in grid size as we must interpolate the value of each uniform grid point across multiple segments to find the optimal value. Moreover, the simple retirement choice problem MSS is applied to features monotone continuation state policies (i.e., the policy function can only jump 'up'). If the continuation state policy is not strictly monotone, then inverting the Euler equation may generate multiple endogenous grid points in the state space for a given point in the continuation state space. In this case, distinct overlapping segments cannot be identified (see p339 in Iskhakov et al. (2017)) since en-

dogenous grid points belonging to distinct segments may be adjacent to each other and increase or decrease from the preceding points along the exogenous grid; detecting where the endogenous grid decreases no longer suffices to separate valid segments from each other. Finally, while Iskhakov et al. (2017) provides a proof for the monotonicity of the continuation state policy function in their example, no theoretical results supply the conditions under which MSS can work.

To illustrate the MSS approach, refer again to the right panel of Figure 2; the set of points $\{(\hat{x}_1, \hat{x}'_1), (\hat{x}_2, \hat{x}'_2), (\hat{x}_3, \hat{x}'_3), (\hat{x}_4, \hat{x}'_4), (\hat{x}_7, \hat{x}'_7), (\hat{x}_9, \hat{x}'_9)\}$ forms a monotone segment when plotted along the vertical axis. When these points are ordered by the exogenous variable $\hat{x}'$ on the exogenous grid, they appear consecutively and are associated with a single future choice-specific policy function. In contrast, the points $\{(\hat{x}_5, \hat{x}'_5), (\hat{x}_6, \hat{x}'_6), (\hat{x}_8, \hat{x}'_8)\}$ belong to a different future choice-specific policy function, which is also monotone over its own domain. MSS exploits the monotonicity of the policy function by performing the following steps:

(i) Order exogenous grid points to identify distinct monotone segments within the exogenous grid by identifying the points such that $\hat{x}_{i+1} > \hat{x}_i$, $x_{i+1} < x_i$ (Fella, 2025).

(ii) Create an interpolant using the current period values over these segments separately.

(iii) Locate the crossing points between these interpolated segments (as illustrated in the left panel of Figure 2).

(iv) Remove candidate points that lie below the dominating upper envelope on either side of each crossing point.

Local triangulation method (LTM). A second approach computes the upper envelope by identifying and comparing local interpolants generated by overlapping future choice-specific value functions (Druedahl and Jørgensen, 2017; Druedahl, 2021).[10] LTM searches the exogenous grid for intervals (i.e., pairs of ordered points) whose corresponding endogenous grid points cover a given point in the uniform state space grid and then takes the maximum of the interpolants intervals. As grid size increases, LTM complexity increases too - quadratically with grid size, since one must loop over the exogenous grid to find intervals for each point in the state-space.[11] Like MSS, LTM requires monotone con-

[10] LTM was introduced by Druedahl and Jørgensen (2017) for DC-EGM problems in multidimensional settings. Note that when applied to one-dimensional settings (Druedahl, 2021), the LTM requirement of policy function invertibility translates to a requirement of strict monotonicity.

[11] The LTM upper envelope routine double loop executes $O(N^{\mathrm{a}} N^{\mathrm{m}})$ operations; when grids are refined together ($N^{\mathrm{a}} \propto N^{\mathrm{m}} \propto K$), workload is $O(K^2)$; see `upperenvelope` in `ConsumptionSaving` (Druedahl, 2024).

tinuation state policies. For LTM to succeed, the intervals it identifies in the exogenous grid must consist of consecutive points that belong to the same future discrete choice sequence. That is, the exogenous grid must be partitioned into disjoint intervals or segments, each of which implies a unique future discrete choice sequence. Druedahl and Jørgensen (2017) acknowledges this and provides results showing the pointwise asymptotic accuracy of their method under invertibility conditions. We show what happens when these conditions break down in Section 2.2.3.

In practical terms, the one-dimensional Druedahl and Jørgensen (2017) procedure reduces to a *pairwise segment comparison* on an interval of the endogenous wealth space. Let $\{\hat{x}_i, \hat{x}'_i, \hat{v}_i\}_{i=1,\dots,N}$ be exogenous- and endogenous-grid pairs as in Figure 2, ordered by the $\hat{x}'_i$ in the exogenous grid. Also fix a uniform grid of current assets $\mathbb{X}_i^U$. The algorithm proceeds as follows:

(i) Identify all consecutive pairs of exogenous points $\left(\hat{x}'_i, \hat{x}'_{i+1}\right)$ in the exogenous grid.

(ii) For each pair $\left(\hat{x}'_i, \hat{x}'_{i+1}\right)$, define a bounding interval $\left[\underline{x}_i, \overline{x}_i\right]$ for current assets x in the endogenous space.

(iii) For each bounding interval $\left[\underline{x}_i, \overline{x}_i\right]$, select those points in $\mathbb{X}^U$ (i.e., a uniform grid of current assets) that lie within the interval. Denote this subset by $\mathbb{X}_i^U$.

(iv) For each point $\hat{x}_j^U \in \mathbb{X}_i^U$, interpolate the value $v_{i,j}^U$ using data from $\left(\hat{x}'_i, \hat{v}_i\right)$ and $\left(\hat{x}'_{i+1}, \hat{v}_{i+1}\right)$. If multiple bounding intervals overlap at $\hat{x}_j^U$, compare the interpolated values $v_{i,j}^U$ across all such intervals i, and select the interval $i^\star$ yielding the highest value.

(v) The pair $\left(\mathbb{X}_i^U, \mathbb{V}_i^U\right)$ serves as an interpolant for approximating the value function. Similarly, an approximate policy function can be constructed by interpolating the optimal policy associated with each $v_{i,j}^U$.

Roof-top cut (RFC). A more recent upper envelope approach is the general multidimensional RFC algorithm proposed in Dobrescu and Shanker (2024). RFC uses the gradient of the future choice-specific value function at each point and removes points below the tangent plane constructed by the gradient. While also able to handle non-monotone policies, RFC still relies on exhaustive nearby nearest neighbor searches to eliminate candidate points, which, in the one-dimensional case, may add significant complexity.

FUES advantage. FUES resolves all the shortcomings of these methods. First, it discards suboptimal points in a single linear pass: a *local secant–slope* test applied to the upper envelope removes dominated candidates, eliminating the need for collecting and

interpolating over all possible segments or covering interval searches. Second, FUES directly tests whether an individual grid point produced by EGM can belong to an upper envelope, rather than construct segments or interpolate over from the exogenous grid. This imposes no restrictions on the structure of exogenous points and admits situations where the policy function is not globally invertible. In addition, unlike any of the methods above, our theoretical results guarantee that FUES recovers the true upper envelope under general conditions.

B Endogenous grid-specific jump detection

In practice, we find that selecting a reasonable constant value for $\bar{M}$ through experimentation (and subsequently verifying the upper envelope is recovered) works well. However, an endogenous jump detection threshold $\bar{M}_i^\star$ can also be theoretically derived based on the curvature of the policy functions, and easily computed for each grid point. To this effect, note that the role of the jump detection threshold is to detect a point $\hat{x}_{i+1}$ that does not lie on the same future choice-specific policy function as $\hat{x}_i$. Consider now that for any real-valued continuously differentiable function f on an interval I, by the Mean Value Theorem, we must have that $|f(y) - f(x)| \leq L|y - x|$, where $L = \sup\{|f'(z)| : z \in I\}$. Let $\sigma_t^{\mathbf{d}}$ be the asset policy function at time t, conditional on a future sequence of discrete choices $\mathbf{d}$ - recall discussion below (5). The functions $a \mapsto \sigma_t^{\mathbf{d}}(a)$ are continuously differentiable and convex (Carroll and Shanker, 2025) as all future discrete choices are held fixed. For any exogenous grid point $\hat{x}_{i+1}$ to imply the same future discrete choices as $\hat{x}_i$, it must be that

$$|\sigma_t^{\mathbf{d}_i}(\hat{x}_i) - \sigma_t^{\mathbf{d}_{i+1}}(\hat{x}_{i+1})| \leq \bar{M}_i^\star |\hat{x}_{i+1} - \hat{x}_i|, \tag{54}$$

where $\bar{M}_i^\star = \max\left\{\left|\partial\sigma^{\mathbf{d}_{i+1}}(\hat{x}_{i+1})\right|, \left|\partial\sigma^{\mathbf{d}_i}(\hat{x}_i)\right|\right\}$ and we use ∂ to denote the derivative of the policy function with respect to financial assets. The term $\mathbf{d}_i$ above is the future sequence of discrete choices implied by the exogenous grid point i. If two points do not satisfy the above inequality, then they cannot lie on the same future choice-specific policy function and we have 'detected a jump' in the policy function.

Starting from the exogenous grid points, we can use the Euler equation for the problem to derive the terms $\partial\sigma^{\mathbf{d}_i}(\hat{x}_i)$ analytically and recursively as follows:

$$\partial\sigma_t^{\mathbf{d}_i}(\hat{x}_i) = \frac{(1+r)u''(\hat{c}_i)}{u''(\hat{c}_i) + \beta(1+r)u''(\hat{c}_i')[(1+r) - \partial\sigma_{t+1}^{\mathbf{d}_i}(\hat{x}_i')]}, \tag{55}$$

where $\hat{c}'_i$ is the next period consumption implied by the exogenous grid point i. It is important to note that the dependence of the policy function derivatives $\partial\sigma_t^{\mathbf{d}_i}$ on the future sequence of discrete choices is *given implicitly* by the exogenous grid point i. Thus, the above derivative can be calculated by evaluating its RHS as follows: first evaluate $u''(\hat{c}_i)$ using the $\hat{c}_i$ implied by the standard EGM step, and then evaluate the next period value $u''(\hat{c}'_i)\left[(1+r) - \partial\sigma_{t+1}^{\mathbf{d}_i}(\hat{x}'_i)\right]$ using an interpolant of the function $u''(c')[(1+r) - \partial\sigma_{t+1}^{\mathbf{d}}(x')]$. In what follows, at a time t iteration, the function $u''(c')\left[(1+r) - \partial\sigma_{t+1}^{\mathbf{d}}(x')\right]$ is evaluated on the exogenous grid points and stored in a grid $\hat{\mathbb{O}}'_t$. (For period T, we can use $\partial\sigma_T^{\mathbf{d}} = 0$ since all wealth is consumed.) In particular, fixing a time t and noting $\hat{o}'_i \in \hat{\mathbb{O}}'_t$, define

$$\hat{\Delta}_i = \frac{(1+r)u''(\hat{c}_i)}{u''(\hat{c}_i) + \beta(1+r)\hat{o}'_i} \approx \partial\sigma_t^{\mathbf{d}_i}(\hat{x}_i), \tag{56}$$

$$\hat{o}_i = u''(\hat{c}_i)\left[(1+r) - \hat{\Delta}_i\right]. \tag{57}$$

Using the above, $\hat{\mathbb{O}}_t$ can be constructed from the points $\hat{o}_i$. The modified pseudo-code for FUES with endogenous jump detection becomes:[12]

Box 4: FUES method with endogenous jump detection

1. Compute $\hat{\mathbb{X}}_t$, $\hat{\mathbb{C}}_t$, $\hat{\mathbb{V}}_t$, $\hat{\mathbb{O}}_t$, $\hat{\mathbb{X}}'_t$, $\hat{\mathbb{C}}'$ and $\hat{\mathbb{O}}'_t$ using standard EGM.
2. Sort all sequences in order of the *endogenous* grid $\hat{\mathbb{X}}_t$.
3. Start from point $i = 2$.
4. Compute $g_i = \frac{\hat{v}_i - \hat{v}_{i-1}}{\hat{x}_i - \hat{x}_{i-1}}$, $g_{i+1} = \frac{\hat{v}_{i+1} - \hat{v}_i}{\hat{x}_{i+1} - \hat{x}_i}$, $\hat{\Delta}_i$, and $\hat{\Delta}_{i+1}$.
5. If the endogenous bound in (54) is violated and a right turn is made ($g_{i+1} < g_i$), then remove point $i+1$ from grids $\hat{\mathbb{X}}_t$, $\hat{\mathbb{V}}_t$, $\hat{\mathbb{O}}_t$, $\hat{\mathbb{X}}'_t$ and $\hat{\mathbb{O}}'_t$. Otherwise, set $i = i+1$.
6. If $i + 1 \leq |\hat{\mathbb{X}}_t|$, then repeat from step 5.

Once the scan is complete for time t, the refined grid $\mathbb{O}_t$ can be interpolated on the refined endogenous grid along with the refined value and policy function. We also note that if the policy functions are Lipschitz bounded and there are only finitely many jumps, then the grid-specific bounds will be bounded and we will have effectively chosen a bound at each grid point such that the jump detection condition in Corollary 1 is satisfied. Hence, for a grid size large enough, FUES will eliminate all suboptimal points without error.

[12]The FUES implementation with endogenous jump detection is provided as an additional option here.

C Intermediate proofs

Proof of Claim 1. Fix any $\epsilon > 0$ such that $\frac{D}{\epsilon} > 2M$ (possible since $D > 0$ and $M < \infty$ by Assumptions 2 and 3).

Let $\hat{\mathbb{X}}'$ be the uniform exogenous grid in X' with $|\hat{\mathbb{X}}'| = N$, and let

$$h_N \;:=\; \max\{\hat{x}'_{i+1} - \hat{x}'_i : \hat{x}'_i, \hat{x}'_{i+1} \in \hat{\mathbb{X}}'\}$$

be its mesh size. Since $\hat{\mathbb{X}}'$ is uniform on a compact interval, $h_N \to 0$ as $N \to \infty$.

By Assumption 4 and finiteness of $\mathbb{D}$, there are only finitely many segments of the interior exogenous grid on which a single discrete choice $\mathbf{d}$ is optimal. On each such segment, the inverse-Euler-equation step of EGM defines a (single-valued) inverse map from $x' \in \operatorname{int} X'$ to $x \in X$. Over the interior interval (i.e., away from the binding constraint), these inverse maps are Lipschitz-bounded with a finite constant - i.e., they satisfy the bounded inverse-Lipschitz (bi-Lipschitz) regularity as in inverse-Euler-equation methods (Dobrescu and Shanker, 2024)). Since there are only finitely many such segments, we may take a uniform bound $M_{\text{inv}} < \infty$ on their Lipschitz constants.

In particular, choose N large enough that $M_{\text{inv}} h_N \leq \epsilon$. Then for any two consecutive exogenous grid points $\hat{x}'_i$ and $\hat{x}'_{i+1}$ belonging to the same optimal interior segment, their inverse images $\hat{x}_i$ and $\hat{x}_{i+1}$ satisfy $|\hat{x}_{i+1} - \hat{x}_i| \leq \epsilon$. Moreover, since the corresponding discrete choice is optimal on that segment, these inverse-image points lie on the upper envelope - i.e., they belong to $\mathbb{X}^\star$.

Finally, when a constraint binds at a bracket boundary, candidate points satisfy $x' = b$ for some $b \in \mathcal{B}$ (including the endpoints $b = \underline{x}' = \inf X'$ and $b = \bar{x}' = \sup X'$, as well as interior bracket boundaries). As described in the main text, we append to the endogenous grid, for each $b \in \mathcal{B}$, a constrained-region grid at $x' = b$ by selecting x points on a uniform grid over the corresponding constrained region. Since we are free to choose the density of these constrained-region grids and $|\mathcal{B}| < \infty$, we may select the densities so that adjacent constrained points are also at most ϵ apart. Combining the interior segments with the boundary constrained-region constructions, for N (and constrained-grid densities) large enough, the ordered set $\mathbb{X}^\star = \{x_0^\star < x_1^\star < \cdots\}$ satisfies $|x_{j+1}^\star - x_j^\star| \leq \epsilon$ for all consecutive pairs, and $\frac{D}{\epsilon} > 2M$ by construction. □

Claim 2. $\mathbb{X}^\star \subseteq \widetilde{\mathbb{X}}^\star$.

Proof. Since each $Q^{\mathbf{d}}$ is concave and $\widetilde{Q}^{\mathbf{d}}$ is its piecewise-linear interpolant, we have $\widetilde{Q}^{\mathbf{d}}(x) \leq Q^{\mathbf{d}}(x)$ for all $x \in X$ and $\mathbf{d} \in \mathbb{D}$, and hence $\widetilde{U}(x) \leq V(x)$ for all $x \in X$. Moreover, for any $\hat{x}_i \in \mathbb{X}^\star$, let $\mathbf{d} \in \mathbb{D}$ be such that $\hat{x}_i \in \mathbb{X}^{\mathbf{d}}$ (so that $\hat{v}_i = Q^{\mathbf{d}}(\hat{x}_i) = \widetilde{Q}^{\mathbf{d}}(\hat{x}_i)$). Then

$$\hat{v}_i \;=\; \widetilde{Q}^{\mathbf{d}}(\hat{x}_i) \;\leq\; \widetilde{U}(\hat{x}_i) \;\leq\; V(\hat{x}_i) \;=\; \hat{v}_i,$$

and thus $\hat{x}_i \in \widetilde{\mathbb{X}}^\star$. In particular, $\mathbb{X}^\star \subseteq \widetilde{\mathbb{X}}^\star$. □

Claim 3. *Let $\mathbb{D}$ be a finite set of future discrete choices, and let X be a compact state space. Suppose that for each $d \in \mathbb{D}$, the region*

$$X^{\mathbf{d}} \;=\; \left\{\, x \in X : \mathbf{d} = \arg\max_{\mathbf{s} \in \mathbb{D}} Q^{\mathbf{s}}(x) \right\}$$

is a non-empty and connected subset of X. Then there exists an integer $N^* > 0$ such that, for all exogenous grids of size $N \geq N^*$, each $X^{\mathbf{d}}$ contains at least two distinct *endogenous grid points.*

Proof. The argument is similar to the proof of Claim 1. □

Claim 4. *Fix indices $i < j < k$ and suppose $\hat{x}_i, \hat{x}_j \in \widetilde{\mathbb{X}}^\star$ and that there exists $\mathbf{m} \in \mathbb{D}$ such that*

$$\hat{v}_i = \widetilde{Q}^{\mathbf{m}}(\hat{x}_i), \qquad \hat{v}_j = \widetilde{Q}^{\mathbf{m}}(\hat{x}_j).$$

Let $\mathbf{q} \in \mathbb{D}$ be such that $\hat{v}_k = \widetilde{Q}^{\mathbf{q}}(\hat{x}_k)$, and assume that at $x = \hat{x}_k$ the interpolant envelope is attained by one of these two interpolants - i.e.,

$$\widetilde{U}(\hat{x}_k) \;=\; \max\{\widetilde{Q}^{\mathbf{m}}(\hat{x}_k),\, \widetilde{Q}^{\mathbf{q}}(\hat{x}_k)\}.$$

If

$$\frac{\hat{v}_j - \hat{v}_i}{\hat{x}_j - \hat{x}_i} \;<\; \frac{\hat{v}_k - \hat{v}_j}{\hat{x}_k - \hat{x}_j}, \tag{58}$$

then $\hat{x}_k \in \widetilde{\mathbb{X}}^\star$.

Proof. Since $\widetilde{Q}^{\mathbf{m}}$ is concave, its secant slopes are weakly decreasing, hence

$$\frac{\widetilde{Q}^{\mathbf{m}}(\hat{x}_j) - \widetilde{Q}^{\mathbf{m}}(\hat{x}_i)}{\hat{x}_j - \hat{x}_i} \;\geq\; \frac{\widetilde{Q}^{\mathbf{m}}(\hat{x}_k) - \widetilde{Q}^{\mathbf{m}}(\hat{x}_j)}{\hat{x}_k - \hat{x}_j}.$$

Substituting $\hat{v}_i = \widetilde{Q}^{\mathbf{m}}(\hat{x}_i)$ and $\hat{v}_j = \widetilde{Q}^{\mathbf{m}}(\hat{x}_j)$ and combining with (58) yields

$$\frac{\widetilde{Q}^{\mathbf{m}}(\hat{x}_k) - \hat{v}_j}{\hat{x}_k - \hat{x}_j} \;<\; \frac{\hat{v}_k - \hat{v}_j}{\hat{x}_k - \hat{x}_j},$$

and hence $\widetilde{Q}^{\mathbf{m}}(\hat{x}_k) < \hat{v}_k = \widetilde{Q}^{\mathbf{q}}(\hat{x}_k)$. Therefore $\widetilde{U}(\hat{x}_k) = \hat{v}_k$ by the two-branch assumption in the statement, so $\hat{x}_k \in \widetilde{\mathbb{X}}^\star$. □

Claim 5. *Consider the setting of Part 1 Case 1 Sub-case (ii) in the proof of Proposition 1. Let x^{TP} be the smallest crossing point in T_P such that $x^{TP} > \hat{x}_{i_k}$. Suppose that $\hat{v}_{i_k} = Q^{\mathbf{m}}(\hat{x}_{i_k})$ and $\hat{v}_{i_{k+1}} = Q^{\hat{\mathbf{m}}}(\hat{x}_{i_{k+1}})$ with $\hat{\mathbf{m}} \neq \mathbf{m}$. If $\hat{x}_{i_{k+1}} \leq x^{TP}$, then $\hat{x}_{i_{k+1}} \notin \mathbb{X}$.*

Proof. Write the endogenous grid post-state policies as

$$\hat{x}'_{i_k} = \sigma^{\mathbf{m}}(\hat{x}_{i_k}), \qquad \hat{x}'_{i_{k+1}} = \sigma^{\hat{\mathbf{m}}}(\hat{x}_{i_{k+1}}).$$

If $\hat{z}_{i_{k+1}} \neq \hat{z}_{i_k}$, then the right-turn jump/bracket-switch test in Algorithm 1 is triggered at line 9 even if $\hat{x}'_{i_{k+1}} = \hat{x}'_{i_k}$ (e.g., along locally flat policies at tax-bracket boundaries). Otherwise, $\hat{z}_{i_{k+1}} = \hat{z}_{i_k}$ and both points lie in the same bracket interval. Set $\Delta x := \left|\hat{x}_{i_{k+1}} - \hat{x}_{i_k}\right|$. By Claim 1, $\Delta x \leq \epsilon$. Using the triangle inequality and then Assumptions 2 and 3,

$$\begin{aligned}
\left|\hat{x}'_{i_{k+1}} - \hat{x}'_{i_k}\right| &= \left|\sigma^{\hat{\mathbf{m}}}(\hat{x}_{i_{k+1}}) - \sigma^{\mathbf{m}}(\hat{x}_{i_k})\right| \\
&\geq \left|\sigma^{\hat{\mathbf{m}}}(\hat{x}_{i_k}) - \sigma^{\mathbf{m}}(\hat{x}_{i_k})\right| \;-\; \left|\sigma^{\hat{\mathbf{m}}}(\hat{x}_{i_{k+1}}) - \sigma^{\hat{\mathbf{m}}}(\hat{x}_{i_k})\right| \\
&\geq D - M\,\Delta x.
\end{aligned}$$

Dividing by Δx and using $\Delta x \leq \epsilon$ gives $\left|\frac{\hat{x}'_{i_{k+1}} - \hat{x}'_{i_k}}{\hat{x}_{i_{k+1}} - \hat{x}_{i_k}}\right| \geq \frac{D}{\epsilon} - M$. Since $\bar{M} < \frac{D}{\epsilon} - M$ (e.g. by choosing $\bar{M} = D/(2\epsilon)$ in Corollary 1), the jump test is triggered at line 9. In either case, the scan flags $\hat{x}_{i_{k+1}}$ at a right turn; since $\hat{x}_{i_{k+1}} \leq x^{\mathrm{TP}}$, the forward scan (if used) cannot certify dominance of $\hat{x}_{i_{k+1}}$ and the point is not retained - i.e., $\hat{x}_{i_{k+1}} \notin \mathbb{X}$. □

D Additional tables and figures

Table 6: Application 1 - Discrete retirement choice model with taste shocks: Speed & accuracy

Grid		FUES		MSS		LTM		RFC	
N	δ	U.Env	E.Err	U.Env	E.Err	U.Env	E.Err	U.Env	E.Err
500	0.25	0.11	-1.19	1.09	-1.17	2.95	-1.17	1.54	-1.16
	0.50	0.11	-1.22	1.21	-1.18	2.97	-1.18	1.41	-1.19
	1.00	0.11	-1.35	0.93	-1.39	2.40	-1.39	1.41	-1.40
1,000	0.25	0.21	-1.19	2.06	-1.17	9.58	-1.17	3.06	-1.13
	0.50	0.21	-1.22	2.16	-1.19	9.30	-1.19	3.20	-1.17
	1.00	0.21	-1.31	1.83	-1.38	7.95	-1.38	2.77	-1.42
2,000	0.25	0.42	-1.20	4.38	-1.16	35.84	-1.16	6.21	-1.13
	0.50	0.42	-1.22	4.24	-1.19	35.11	-1.19	6.70	-1.17
	1.00	0.42	-1.31	3.48	-1.38	32.13	-1.38	6.43	-1.42
3,000	0.25	0.64	-1.20	7.08	-1.15	78.92	-1.15	10.88	-1.13
	0.50	0.64	-1.22	6.98	-1.20	78.36	-1.20	12.82	-1.18
	1.00	0.65	-1.29	4.99	-1.38	72.74	-1.38	12.83	-1.41

Notes: Parameters: $T = 50$, $\beta = 0.96$, $r = 0.02$, $y = 20$, $\bar{s} = 0.05$, $a_t \in [0, 500]$. **U.Env**: average upper envelope time per period. **E.Err**: Euler equation error in mean $\log_{10}$ deviation. All time in milliseconds.

Table 7: Application 3 - Discrete housing choice model (homeowner problem): Speed & accuracy

Grid		FUES				MSS				LTM				VFI		
W	H	U.env	Cons.	Total	E.Err	U.env	Cons.	Total	E.Err	U.env	Cons.	Total	E.Err	Cons.	Total	E.Err
1,000	3	0.01	0.01	0.23	-5.79	0.03	0.04	0.24	-5.78	0.20	0.20	0.39	-5.78	0.82	1.01	-3.46
	5	0.01	0.02	0.25	-5.75	0.06	0.07	0.27	-5.75	0.33	0.34	0.53	-5.76	1.25	1.44	-3.46
	7	0.01	0.02	0.24	-5.75	0.08	0.09	0.30	-5.74	0.46	0.47	0.66	-5.76	1.62	1.81	-3.46
2,000	3	0.01	0.01	0.24	-6.34	0.06	0.07	0.28	-6.33	0.80	0.81	0.99	-6.33	1.48	1.66	-3.75
	5	0.01	0.02	0.25	-6.28	0.13	0.14	0.36	-6.28	1.32	1.34	1.53	-6.29	2.37	2.55	-3.75
	7	0.01	0.03	0.27	-6.29	0.18	0.20	0.42	-6.26	1.81	1.83	2.03	-6.28	3.09	3.29	-3.74
4,000	3	0.01	0.02	0.25	-6.85	0.15	0.17	0.38	-6.84	3.31	3.32	3.50	-6.85	2.87	3.06	-4.05
	5	0.02	0.04	0.29	-6.78	0.31	0.33	0.56	-6.76	5.51	5.53	5.73	-6.77	4.77	4.97	-4.03
	7	0.02	0.05	0.32	-6.78	0.45	0.48	0.74	-6.76	7.19	7.21	7.43	-6.77	6.19	6.41	-4.03
6,000	5	0.02	0.05	0.28	-7.11	0.48	0.50	0.72	-7.09	12.03	12.03	12.23	-7.10	6.71	6.90	-4.21
	7	0.03	0.06	0.31	-7.09	0.72	0.74	0.97	-7.06	16.61	16.61	16.81	-7.08	8.98	9.18	-4.20
	10	0.04	0.09	0.34	-7.06	1.11	1.13	1.38	-7.04	23.36	23.36	23.58	-7.06	12.23	12.44	-4.20
8,000	5	0.03	0.06	0.29	-7.32	0.67	0.69	0.91	-7.30	21.38	21.38	21.57	-7.31	8.81	9.00	-4.33
	7	0.03	0.08	0.32	-7.30	1.02	1.03	1.26	-7.28	29.50	29.50	29.70	-7.29	11.79	11.99	-4.32
	10	0.04	0.10	0.36	-7.28	1.57	1.57	1.82	-7.25	41.49	41.49	41.71	-7.27	16.05	16.26	-4.32
10,000	5	0.03	0.06	0.30	-7.49	0.88	0.89	1.11	-7.47	33.38	33.38	33.57	-7.48	10.88	11.07	-4.42
	7	0.04	0.09	0.33	-7.47	1.33	1.33	1.56	-7.44	46.06	46.06	46.26	-7.46	14.56	14.76	-4.42
	10	0.05	0.12	0.38	-7.44	2.03	2.03	2.27	-7.42	64.79	64.79	65.01	-7.44	19.82	20.03	-4.41

Notes: Parameters: $r = 0.06$, $\beta = 0.93$, $T = 20$, $\alpha = 0.77$, $\tau = 0.07$, $\kappa = 0.075$, $\iota = 0.01$, $A_{\max} = 40$, $H_{\max} = 5$, $w_{\max} = 40$, $P^r = 0.1$, $\vartheta = 0.5$. Income: $\log y = z + \varepsilon$; $z_t = \rho z_{t-1} + \eta_t$ with $\rho = 0.977$, $\sigma_\eta = 0.024$, $\sigma_\varepsilon = 0.063$ on 7×7 grid. **U.Env**: average upper envelope time per period. **Cons.**: average homeowner consumption stage time per period. **Total**: average total solution time per period. **E.Err**: Euler equation error in mean $\log_{10}$ deviation. All time in seconds.

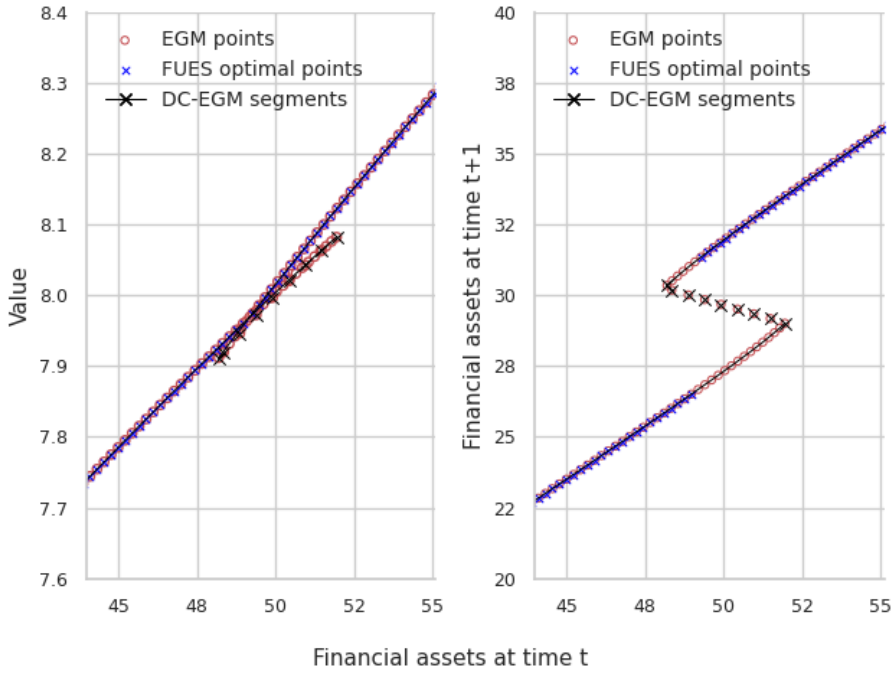


Figure 7: Value correspondence and optimal points for $t = 17$ with taste shocks in Application 1.

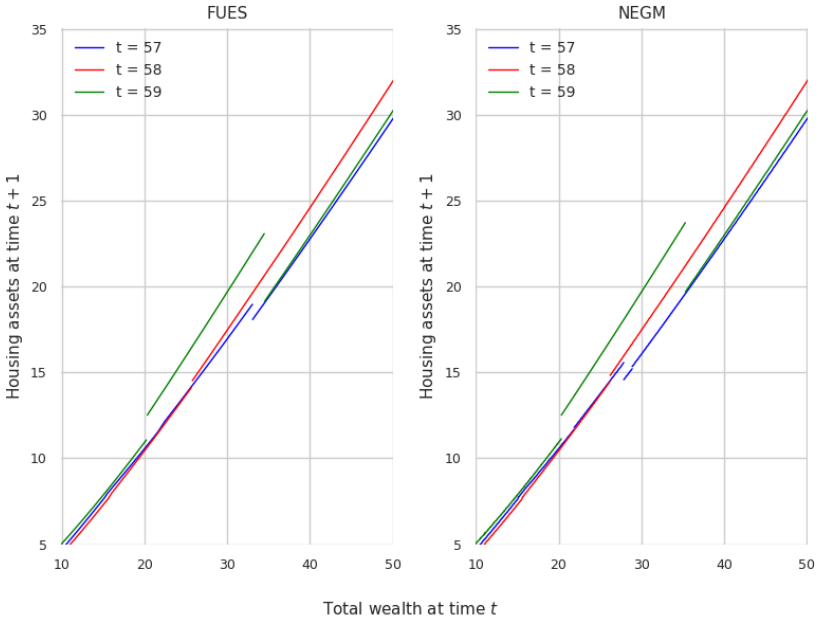


Figure 8: Housing policy functions computed using FUES and NEGM in Application 2. Financial assets grid size is set to 600, as is for the housing assets grid.

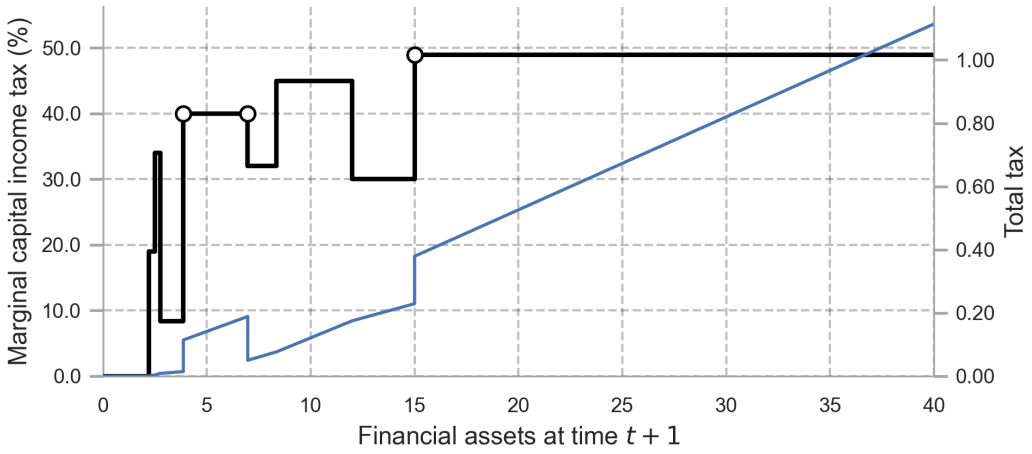


Figure 9: Piecewise-linear capital income tax schedule in Application 3. Open circles indicate jump discontinuities.